Part B: technical description

ERNEST - EUROPEAN RESEARCHERS NIGHT ESCAPE SCIENCE TEAM

#@APP-FORM-HECSA@#

List of participants

Participant No. *	Participant organisation name	Country
1 (Coordinator)	Istituto Nazionale di Fisica Nucleare - **INFN**	Italy
2	Istituto Nazionale di Astrofisica - **INAF**	Italy
3	Consiglio Nazionale delle Ricerche - **CNR**	Italy
4	Centre National de la Recherche Scientifique CNRS - **CNRS**	France
5	Università degli Studi di Bari Aldo Moro - **UniBa**	Italy
6	Università di Pisa - **UniPi**	Italy
7	Università di Pavia - **UniPv**	Italy
8	Università degli Studi di Perugia - **UniPg**	Italy
9	The University of Manchester - **UniMan**	UK
10	Università del Piemonte Orientale - **UniUPO**	Italy
11	Università degli Studi di Torino - **UniTo**	Italy
12	LAMA Società Cooperativa Impresa Sociale - **LAMA**	Italy

1. Excellence #@REL-EVA-RE@#

The **ERNEST (European Researchers Night Escape Science Team)** project aims to establish a European (EU) network of institutions and scientists using scientific educational escape rooms - where participants learn through interactive real-life puzzle solving in immersive environments - to engage citizens with science and stimulate curiosity about careers and research in STEM fields.

1.1 Objectives #@PRJ-OBJ-PO@#

Specific objectives of the ERNEST projects are:

1. Network creation: The project will connect European institutions active/interested in using escape rooms for citizen engagement. A dedicated website will serve as a hub for network members, resources, best practices, and event promotion, ensuring broad visibility and collaboration across Europe.

2. Co-creation of science escape rooms: The ERNEST members will design, test, and implement innovative, scientific escape room scenarios and games, targeting different subjects, age groups and interests. The partners' prior experiences on scientific escape rooms and game development (see section 1.2.4.2) will provide the foundation for co-creation, exploiting the different competencies and capacity building. Particularly in the context of WP3 this aspect will be explored in collaboration with school teachers.

3. Public Engagement and Inclusivity: ERNEST will engage diverse audiences (youth, families, students, science enthusiasts as well as the wider public, with particular emphasis on underserved audiences) through accessible, inquiry-based, enjoyable experiences. Through the instrument of our guided escape rooms, we also aim to overcome the use of overextended masculine, typical of our languages, for a more balanced science communication. The partners' experience with students with special needs will ensure further inclusivity in all activities. In addition, the Researchers at School phase of the project will allow ERNEST to reach also smaller towns, rural areas and remote parts of the country, presenting the scientific escape rooms also to students in these areas thus striving to achieve equal opportunities of exposure.

4. Impact Evaluation and Dissemination: Using pre- and post-activity surveys, the project will measure knowledge, engagement, and skill development, investigating any changes in people's perception of scientists and the research process as a possible outcome of the project. The gamified nature of escape rooms allows for both qualitative and quantitative data collection, providing a unique setting to test science engagement methods. These insights will be shared on the project website, along with reports for science education stakeholders.

5. Dissemination and communication of ERNEST material: To further the goals of European Open Science, all material produced by the ERNEST network will be made available following the FAIR principles (https://www.go-fair.org/fair-principles/) and advertised in conferences and network meetings, so that the escape rooms are easily

reproducible by anyone in the world (exercise already done by UniMan for the original INFN HEPscape! room). In addition, simpler game setups will be made available for science teachers to download from the ERNEST web server on a diverse set of scientific subjects, including walk-through instructions of the scientific content to set up games at their schools.

1.1.1 Relation to the work program

ERNEST directly addresses the objectives of the **MSCA CITIZENS** call:

1. Bringing research closer to the public: ERNEST uses scientific escape rooms as innovative outreach tools, enabling citizens to engage actively with science and meet researchers in informal settings.

2. Raising awareness of the impact of researchers work on citizens daily lives: Escape rooms will feature challenges tied to societal issues and UN Sustainable Development Goals (SDG, e.g., good health and well-being, gender equality, climate action), showing how research addresses real-world problems.

3. **Increasing citizens' understanding of science and encouraging young people to embark on research careers:** The playful, inquiry-based format will stimulate curiosity and deeper understanding, particularly among young people. By collaborating with schools, the project will **reach students in a format that encourages exploration and demystifies scientific careers**.

4. Promoting open science practices through co-creation and dissemination of methods and materials: All materials will be openly shared, enabling broad adoption, replication, and scaling across Europe for lasting impact. This will enable other institutions to replicate the methodology, maximizing the project's long-term impact.

5. Supporting gender balance in STEM: At least 50% of researchers in ERNEST events and school activities will be women, highlighted as role models in escape rooms, talks, and digital content. Gender-disaggregated data will track girls' and young women's participation in schools and outreach events. A dedicated expertise on this aspect will be represented within the consortium by LAMA.

1.1.2. Measurability, Verifiability, and Realistic Achievability of ERNEST Objectives

All ERNEST objectives are operationalized through clear indicators, systematic data collection, and established partner expertise. Table 1.1 outlines the key objectives of the ERNEST initiative, along with corresponding Key Performance Indicator (KPIs) and target metrics or verification methods to assess progress and impact.

Table 1.1: ERNEST key objectives and corresponding KPIs, verification and achievability targets.

Objective	KPIs	Target / Verification	Achievability
Network Creation	- Creation of **website** - Website engagement - Number of **institutions** joining and engaged	- ≥ 15 institutions joining; ≥ 10 actively participating - ≥ 1000 site visitors/year; ≥ 300 downloads of resources/year	Partners' prior experience with networks and collaborations within the community, providing a solid starting point for mapping and recruitment. This ensures that scenario co-creation and testing are realistic within the project duration.
Co-creation of Escape Rooms	- Number of **scientific topics and scenarios** tested - **Researcher and citizen** involvement	- ≥ 5 new/adapted topics - ≥ 5 new/adapted scenarios - ≥ 50 researchers and ≥ 100 teachers involved	
Public Engagement	- Number of **escape room sessions** delivered - **Audience** reached - **Inclusivity measures** implemented	- ≥ 50 sessions across Europe - ≥ 4000 participants/year - Adaptations for at least 3 target groups - Translation in ≥ 3 languages	Inclusivity refined from previous experience (e.g., deaf students and learners with special needs). Consolidated network of local schools for a lasting impact on the territory.
Impact Evaluation and Dissemination	- Pre- and post-activity **questionnaires**, observation grids, and feedback tools - **Data** on engagement - **Open-access resources** - **Social media** engagement and **talks** at conferences	- 85% sessions evaluated - Data from ≥ 3,000 users - ≥ 10 resources online - ≥ 3 conferences; ≥ 2 policy briefs - Tracking activity and downloads on GitHub and Zenodo pages	Commitment to open-access practices make systematic evaluation and wide dissemination more feasible and realistic
Alignment with the work program	- **Researcher involvement** in activities - **Youth participation**	- ≥ 50 researchers facilitating - ≥ 2000 young participants - 100% scenarios linked to societal	Historical participation of ERNEST partners in ERN networks and the long-standing

	- Escape rooms themes linked to **societal challenges**	issues	reliability of the institutions will ensure a success of the project

#§PRJ-OBJ-PO§#

1.2 Coordination and/or support measures and methodology #@CON-MET-CM@# #@COM-PLE-CP@#

1.2.1 ERNEST main idea: vision and format

The **ERNEST project uses scientific escape rooms to engage citizens**, share research, and inspire interest in science careers. A **European network of institutions** will collaboratively design and evaluate experiences linking science, challenges, and societal issues. The project will **create scientific accurate escape room scenarios** that include puzzles and narrative elements grounded in real-world research and contemporary scientific questions, encouraging participants to think critically, collaborate, and engage directly with scientific methods and reasoning.
To ensure scalability, visibility, and sustainability, ERNEST will **establish a dedicated online platform** serving as a central hub for the network, hosting open-access resources, facilitation guides, co-created scenarios and evaluation tools. An integral component of the methodology is a **robust, embedded evaluation framework** to assess the impact of the escape room experiences on scientific literacy, curiosity, engagement, perception of scientists and attitudes towards science and research careers. Unlike traditional European Researchers Night (**ERN)** projects, which are often organized as regional events concentrated in centralized locations and aimed at gathering local audiences, **ERNEST proposes a distributed, thematic format** driven by a unifying narrative: the **science-based escape room**. Thanks to the **portable, modular nature** of our escape room kits, we can implement engaging, hands-on science activities in a wide range of **unconventional venues**—including classrooms, museums, libraries, historical buildings, public squares, and temporary structures such as gazebos. This flexibility allows ERNEST partners to plan outreach activities well in advance and bring ERN events directly into the **heart of urban life**, activating **locations that are typically not covered** by existing ERN initiatives but are **geographically and culturally close to the partner institutions**. The same approach can be applied to researcher-led school activities, allowing them to reach non-traditional educational settings and bring the ERNEST atmosphere and objectives to schools and **locations that have rarely been exposed to ERN activities or engaged before**. This format addresses existing gaps by offering immersive, gamified science communication that can engage diverse audiences deeply.

1.2.2 Topics considered and disciplines involved

The network collectively covers all **Horizon Europe Missions** with strong, complementary expertise. In **climate-related areas**, **CNR**, **INFN**, **UniTo**, **UniPi** and **UniMan** contribute to renewable energy, urban sustainability and atmospheric physics, while **CNR**, **UniPg**, **UniBa** and **UniMan** address climate adaptation through environmental risk assessment and Earth observation. **Health and cancer research** is represented through extensive activity in biomedical physics and health technology innovation from **INFN**, **UniPv**, **UniTo**, **UniPi**, **CNR** and **UniUPO**. **CNR**, **UniBa**, **UniPg** and **UniTo** work across **soil health**, **food systems** and **biodiversity**, and **CNR**, **UniBa** and **INFN** contribute to **ocean and water monitoring**. **Space and astrophysics** are well represented by **INAF**, **INFN**, **CNRS**, **UniMan**, **CNR** and **UniPg** with strengths **in planetary science, gravitational waves and Earth observation**. **Digital transformation** is a transversal capability — **INFN**, **UniTo**, **UniPi**, **UniMan**, **CNR**, **INAF**, **CNRS** and **UniUPO** are active in **AI**, **data science** and **high-performance computing**. Finally, all partners share a commitment to open science, public engagement and educational innovation. The Steering Committee (as detailed in WP5 Tab 3.1b) will ensure that all participating researchers recognize ERNEST as a unique platform to connect with the public, highlighting how various research centers and universities work together to advance scientific knowledge.

1.2.3 Stakeholders and their engagement

The ERNEST project will bring together a wide network of stakeholders, each contributing specific expertise and audiences to ensure both reach and long-term sustainability. These actors operate at the **local, national, and international** level and represent key categories such as **education (E), research (R), culture (C), media (M), policy (P), and the private sector (PS)**. Their engagement is carefully mapped to the project's WPs, ensuring that activities are inclusive and well-integrated (see Table 1.2).

Table 1.2: Summary of key stakeholder groups, with areas of influence, and specific examples of engagement.

Stakeholder	Short description	Type	Category	Related WP(s)	Examples
Schools & Teachers	Educators engaged in	Local / National	E	WP2, WP3,	E.g. Liceo Ruffini Viterbo, Liceo Majorana Termoli, Liceo Respighi Piacenza, ISISS

	training to design and run scientific escape rooms			WP4	Magarotto Rome, Liceo Tullio Levi-Civita Rome, I.C. Massari-Galilei Bari, ICS "G.Mazzini" Bari, Erasmus+ networks, local teacher associations, Organisers of Science Festival in Manchester (e.g. Universally Manchester festival)
Students	Puzzle co-creators at different levels	Local / National	E	WP2, WP3, WP4	AISF, primary/middle schools, Formazione scuola-lavoro programs at High Schools, student clubs, scouting associations in the UK, PONYS (https://www.na.infn.it/divulgazione/ponys)
Education Boards & Authorities	Support alignment with curricula	National	E	WP1, WP3, WP5	AIF, PLS, Ministries of Education, regional school boards, education offices
Research Inst. & Universities& National Laboratories	Provide scientific expertise and facilitators	Local / National / International	R	WP1, WP2, WP3, WP4	Universities, COST Actions (EPIGRAPHY), SIF, EPS, IPPOG, Fondazione Lincei Scuola, Cockcroft Institute of Accelerator Science, Photon Science Institute (Manchester). National labs (e.g. Daresbury Laboratory), Italian Association of Medical Physics (Italy)
Science Museums & Cultural Centres and Events	Host ERNEST events and attract broad audiences	Local / National	R&C	WP1, WP2, WP3	Salone Internazionale del libro di Torino , Play (Modena- Bologna), Focus Live (Milano), Festival della scienza di Genova, Festival della comunicazione di Camogli, Lucca Comics& Games, Festival delle scienze di Roma, Museo dei bambini di Roma, Città della scienza (Napoli), Comicon (Napoli), Partenoplay (Napoli), Parla Potabile (Napoli), Etna Comics Catania, Science and Industry Museum (Manchester), Fête de la science (https://www.fetedelascience.fr/),"Pole Égalité des Chances" a Department of the École Polytechnique
Libraries & Community Centres	Accessible venues for underserved audiences	Local	R&C	WP1, WP2, WP3	Centro per le famiglie (Piacenza), Teatro Splendor Aosta, Ludoteca CittadiNA, charities in the Greater Manchester area, Public libraries, municipal centres, Youth hubs
NGOs & Youth Associations	Ensure inclusivity	Local / National	R&C	WP1, WP2, WP3	Amnesty Youth, Scouts (Italy and UK), local STEM NGOs
Press & Media Outlets	Disseminate ERNEST	Local / National	M&C	WP1, WP2, WP4	La Repubblica, Rai Scuola, Euronews Science, local TV/radio, Communication services of École Polytechnique, IN2P3 and CNRS and INFN. University News and Media Relations team
Science Communicators & Influencers	Reach digital-native audiences	National / International	M&C	WP1, WP2, WP3	YouTube STEM creators, TikTok and Instagram influencers, science podcasts
Local & Regional Authorities	Provide logistical and political support for ERN events	Local / National	C&P	WP1, WP2, WP5	Città di Roma, Viterbo, Termoli, Napoli, Catania, Bari, Pisa, Pavia, Perugia, Torino, Novara. Regione Valle D'Aosta, Piemonte, Lombardia, Toscana, Umbria, Lazio, Campania, Puglia, Sicilia and their educational offices (uffici scolastici regionali). Policy engagement institute Policy@Manchester, STFC public engagement team. Large multi-academy trusts (e.g. Laurus Trust).
EU	Showcase	International	C&P	WP1,	EU Parliament, DG EAC, MSCA Citizen officers

Policymakers & Repres.	ERNEST as model at Europe level			WP2, WP5	
Private Sponsors & Partners	Contribute resources for sustainability and outreach	National / International	PS	WP1, WP2, WP5	Tech companies, publishers, banks/foundations, Paleos formazione docenti, European Conference on Games Based Learning, Escape rooms industries (e.g. Escape4Change, https://escape4change.com/)

1.2.4 Overall methodology, including the concepts, models and assumptions that underpin the project and how this will enable the network to deliver project's objectives

ERNESTs transform **real scientific research** into **hands-on challenges** aligned with **Horizon Europe Missions**. Designed for **inclusivity**, our escape rooms engage diverse learners through **multisensory**, **collaborative puzzles** that blend **physical and digital elements**. Their **flexible format** allows seamless adaptation across **schools**, **museums**, **festivals**, and a wide range of **audiences**, including **youth**, **underserved communities**, and **special-needs groups**.

1.2.4.1 Evidence-based rationale for using escape rooms in science education

ERNEST is based in robust research evidence and field experience showing that gamified, inquiry-driven methods like **scientific escape rooms effectively promote science engagement, spark curiosity, and cultivate critical thinking and teamwork across diverse groups**. The scientific escape rooms will exploit the 4C learning methodology: Critical thinking, Cooperation, Creativity, Communication *(Creative Education, Vol.6 No.2, Feb17, 2015)*. Research has shown that Game Based Learning increases engagement and retention in STEM education *(Deterding et al., 2011; Hamari et al., 2014)*, while escape rooms specifically have demonstrated effectiveness in advancing problem-solving skills, collaboration, and conceptual understanding *(Veldkamp et al., 2020; Clarke et al., 2017)*. These immersive environments allow participants to interact with scientific challenges in authentic contexts, actively applying scientific reasoning rather than passively consuming information. This aligns with the principles of experiential learning *(Kolb, 1984)* and has been shown to support the development of transversal skills essential for lifelong learning *(Fotaris & Mastoras, 2019)*. In the context of citizen science, the co-creative and participatory nature of escape rooms empowers citizens to engage with real-world research questions and interact directly with researchers, supporting open science practices and building trust in science *(Bonney et al., 2016)*.

1.2.4.2 Existing scientific escape rooms within the network

Several partners within the ERNEST network have already developed and successfully implemented scientific educational escape rooms, demonstrating the feasibility and effectiveness of this methodology for engaging citizens in science. These existing projects serve as **foundational models for the ERNEST activities** and will constitute the basis for the **development of new, innovative setups**, described in Sec 1.2.4.3. During school-based implementations, **our teams have worked iteratively with teachers** to adapt the design, puzzles, and facilitation of escape rooms to accommodate diverse learning needs and accessibility requirements, ensuring that activities are engaging, understandable, playful and participatory for all learners. This **iterative, user-centered approach** will continue within the project. Below are some key examples of scientific escape rooms created by partners:

1. HEPscape! (INFN, https://web.infn.it/hepscape/) is an innovative educational escape room concept designed to immerse participants in the world of high-energy physics (HEP). It is built in the form of a **portable kit** that can be installed in diverse locations like schools, science fairs, museums and festivals. In HEPscape! teams of participants enter a **themed room equipped with scientific props**, experimental replicas, and puzzle-based challenges **rooted in real HEP research at the Large Hadron Collider (LHC) experimental area.** The storyline connects all tasks into a coherent mission—mirroring the process of a physics experiment from hypothesis to discovery of a new particle, the Higgs boson! The storytelling and the riddles can be **adapted to the age of the audience**. Each experience lasts roughly 40 minutes and hosts about 20 people (depending on the dimensions of the room). Since its launch in 2021, HEPscape! has attracted more than 10k visitors, ranging from school groups to public audiences. It has been presented at major science events including Festival della Scienza di Genova, Futuro Remoto, National Geographic Festival delle Scienze in Rome, Lucca Comics&Games and ERN events (NET and SHARPER) in many Italian cities like Rome, Perugia, Bari, Padova, Catania. A dedicated session **in the language of signs** with a Rome school (ISISS Magarotto) was also successfully performed. Feedback surveys indicate a high level of engagement, with participants reporting increased enthusiasm for scientific subjects and improved understanding of scientific processes.

2. A cavallo di un fotone (INAF, https://play.inaf.it/escape-room/) explores the nature of **light,** which **travels across space** carrying information about **celestial bodies, in an interactive, gamified format.** This escape room

introduces participants to the fascinating world of astrophysics, engaging them with robotics and unplugged coding challenges that encourage the practice of computational thinking**. Through a series of puzzles and challenges, participants learn about the nuclear fusion reactions that take place at the center of the Sun, the internal structure of our star, the light travel time across the Solar System, the properties of different planets and the major telescopes that astrophysicists use on Earth to investigate the cosmos**. This escape room has received significant recognition at national science festivals (Festival della Scienza di Genova, 2022; Focus Live, 2024) for its educational value and ability to engage audiences in complex scientific topics in an entertaining way.

3. A caccia nel buio escape room (**INAF**, https://l.infn.it/escape) explores the **formation of stars** and planets in the universe. The escape room revolves around the discovery of a planetary system using images from the Atacama Large Millimeter/submillimeter Array (ALMA), a major radio telescope located in the Chilean Andes. The chosen setting is the researcher's office, who is a woman in astronomy, enabling facilitators to also tackle the **topic of gender issues in science**. Participants are divided in teams, decoding a piece of the scientist's message that will reveal her discovery to the world. The experience allows participants to discover not only the topic of stellar and planetary formation, but also to experience, in a light and playful way, the discovery process and work dynamics behind scientific research.

4. Alla Ricerca dell'Equilibrio Perduto (**CNR**, https://iris.cnr.it/handle/20.500.14243/433033) The game is based on a storyline: a scientist, now unconscious, has hidden the clues to access a password protected box. The activity is thought up, realized, conducted by several CNR institutes in Campania committed to scientific dissemination: IAC, IGB, INO, IPCB, IPSP, IRET, ISASI, SPIN. Every Institute provided a puzzle or small experiment in mathematics, physics, chemistry, and biology, based on their activity/expertise/outreach activity. In this way, many different disciplines are explored. But the components to solve them are also hidden in the room. To solve the puzzle, no prior knowledge is required (apart from reading and logical thinking), as all the "theory" is provided through infographics, explanations, and text to be read. The materials –also made from common use objects- are hidden amongst "distractors" and must be connected in the "right" way. The sequence is modular: some puzzles can be added or removed to increase or decrease the difficulty and to be adapted to different public categories and is "re-designed" in every "representation".

5. Medical Physics Escape Room (UniTo/INFN, https://fisica61.wordpress.com/chi-siamo/) has been designed in **collaboration with the Italian Association of Medical Physics** to raise interest in the general public about the application of Physics in Medicine, encouraging at the same time problem-solving, teamwork, and lateral thinking (fully included in the EU education policies, such as "Key Competences for Lifelong learning"). Three different environments have been designed, focusing on three fields of application of Physics in Medicine: cancer diagnosis, cancer treatment and research (mainly on innovative technologies for beam monitoring in charged particle therapy, i.e. the Torino research activities). Small groups of players are guided by a researcher or a medical physicist, who helps participants learning through playing. This experience has shown that an Escape Room is an excellent tool to communicate to the population the role of the medical physicist in research and clinical practice. This first attempt, addressing the general public, is now available to be used as an off the-shelf tool, but could be modified and improved over specific areas of content, potentially becoming an innovative medical physics educational activity for medical physicists, radiology residents, or other categories.

6. maSTEM game (**UniTo,** https://www.mastem-game.unito.it/) integrates **STEM** challenges in a **playful**, **inquiry-based** format aimed at young audiences. This escape room incorporates interactive challenges from the realms of **physics applied to medicine, biology, veterinary sciences, and mathematics** with multilingual and CLIL (Content and Language Integrated Learning) methodologies. CLIL offers a dual focus on both disciplinary content and language development, making it particularly effective in engaging learners cognitively and linguistically. In maSTEM Game, scientific concepts are explored through English or other foreign languages, mirroring the authentic multilingual environment of contemporary science, where English has become the lingua franca of research, innovation, and collaboration. By linking STEM challenges with linguistic mediation, the project promotes a deeper conceptual understanding, encourages communication and reasoning skills, and fosters intercultural competence. CLIL also supports inclusion: it values diverse linguistic backgrounds and allows non-native speakers to engage with science on equal footing through cooperative tasks.

7."PER me si va ne la fisica recente" (**UniPv**, https://fisicapaviaeducational.it/phys4teens/p4t_per/, Particle Escape Room) was developed by the outreach group "Physics4Teenagers", based at the Department of Physics "Alessandro Volta". **Inspired by Dante's Divina Commedia**, the escape room follows a narrative journey led by Democritus, who takes on the role of Virgil, guiding participants through key experiments and ideas that led to our current understanding of fundamental particles. The storyline concludes with the character **Curiosity, inspired by Beatrice, who accompanies participants beyond the Standard Model of particle physics** (SM), toward open questions such as dark matter, neutrino oscillations, and fundamental force hierarchy. Players progress through a series of puzzles, each corresponding to a pivotal scientific discovery or experiment. As the puzzles are solved, participants gradually gather all the pieces to complete a "puzzle" that represents the SM. Importantly, each step is

followed by a debriefing session to discuss and reinforce the underlying physics concepts. The puzzles are mostly constructed from affordable, recycled or easily available materials, and are intentionally designed to be modular and transportable.

8. "Acceler8 Escape" (**UniMan**, Cockcroft Institute) places participants inside a particle accelerator, working in teams to **solve 8 interactive puzzles corresponding to key elements of an accelerator** - lasers, photocathodes, accelerating cavities, magnets, detectors, vacuum, radiation and safety. Each puzzle is directly linked to the real science using a **combination of visual, audio and physical problem-solving** that reveal a code to unlock a customized stamp for an "escape passport", providing an immersive experience and take-home material that strengthens participant engagement and learning. The activity is designed to last approximately 45-60 mins in total and can accommodate up to 60 participants (2 groups of 30) through non-linear open gameplay across the 8 puzzle stations, with an opening and closing debrief to reinforce key learning messages. Acceler8 Escape has been **developed in partnership with local school staff experienced in both commercial and educational escape room design and delivery**, enabling the creation of authentic puzzles that can be tailored to any age or experience level. It is **supported by the Science and Technology Facilities Council (STFC) public engagement team** at Daresbury Laboratory and officially launched in October 2025 at the Daresbury Family Science Day, welcoming up to 500 participants from local socio-economically deprived communities.

The collaborative nature of the ERNEST network will facilitate knowledge sharing among partners, allowing for the adaptation and expansion of these models into new contexts and disciplines.

1.2.4.3 Scientific Inspiration for new escape rooms

We aim to develop new escape rooms scenarios inspired by a broad spectrum of scientific domains represented across the network, with the following thematic directions:

1. Gravitational waves: These open a narrative space around the detection of cosmic events—such as black hole mergers—introducing concepts like spacetime, interferometry, and signal extraction in an accessible way. This topic will be closely linked to the upcoming **Einstein Telescope project**, a cutting-edge scientific initiative set to be developed in Europe in the coming years. Aligning with this major experimental effort, this escape room will offer a timely opportunity to engage the European public with a project that is poised to place Europe at the forefront of global scientific research.

2. Neutrino telescopes and deep-sea science (inspired by the KM3NeT experiment): Imagine stepping into an underwater lab where the ocean's darkness hides both fragile ecosystems and cutting-edge detectors. Players become scientists racing to trace a mysterious neutrino signal from deep space. To solve the mission, they must crack puzzles inspired by particle physics, overcome challenges of deep-sea engineering, and protect the marine environment—all while uncovering the story behind one of the universe's most elusive particles.

3. Quantum technologies: "Entangled Escape": an educational escape room that immerses participants in the world of quantum physics and quantum optics, transforming counterintuitive principles such as superposition, entanglement, interference, and quantum error correction into interactive puzzles. Players enter a quantum research lab on the verge of decoherence and must work together to stabilize the experiment before time runs out. Each challenge is inspired by simplified experiments that mirror real laboratory practices—for example, aligning polarizers to reveal hidden information illustrates how photons are used to probe polarization states.

4. Real-time analysis in high energy physics and industry: this escape room theme, developed within the outreach hackathons of the MSCA Doctoral Network SMARTHEP, focuses on the vast amount of data produced by Big Science and the need for rapid decision-making aided by novel tools such as AI and modern computing architectures. In a fast-paced environment, the participants solve puzzles to understand the volume of data at the LHC, how the real-time decision making (trigger) system works through sorting games, and how challenging it is to make new discoveries. As part of this project new room ideas on general sustainability theme play a crucial role (two areas of exploration could be are compact particle accelerators and energy efficient data storage).

5. The Turtle Mission: a mission to protect and stand for the planet, facing the challenges related to sustainability, waste management, biodiversity. Players discover an abandoned genetic engineering lab where they could reconstruct a prototypical synthetic organism, a giant turtle that fed plastic and degraded it in a sustainable way. The different tasks involve chemistry, biology, and physics.

6. The search for extraterrestrial intelligence: Astronomers sitting at a radio telescope are performing their routine observations of some stars known to host planets, when suddenly, a mysterious message is detected in the data. The content of the message does not match any known physical mechanism in the source – a well-known star located around 50 light years away. After closer inspection, the puzzling code appears to have been sent by an extraterrestrial civilization. The challenge entails decoding the message and interpreting its meaning, using hints present in the room while also practicing digital skills and computational thinking and questioning, at the same time, our anthropocentric view around the question "is there life elsewhere in the universe?".

Each of these topics provides not only an **engaging narrative hook** but also a chance to promote scientific literacy, **highlight current European research**, and show the direct **relevance of science to society and the future**. Our plan is to use the **same kit adaptable to multiple scenarios** (escape room setups), allowing us to share instruments, tools, and equipment efficiently across different formats.

1.2.4.4 Key concepts and models

The project methodology is underpinned by the following key concepts and models:

- **Experiential learning and gamification**: Promote active participation, problem-solving, and storytelling for deeper learner engagement.
- **STEM and CLIL interplay**: Combine subject knowledge and foreign language skills to activate language and scientific reasoning, enhancing memory, motivation, and creativity.
- **Co-creation and open science: Researchers, educators collaboratively design scenarios ensuring scientific accuracy, pedagogical effectiveness, and alignment with societal challenges.**
- **Iterative design and user-centered adaptation**: Incorporate continuous feedback to adapt experiences for diverse audiences, including underserved learners.
- **Impact evaluation embedded in experience**: Use qualitative and quantitative tools to measure scientific literacy, engagement, and transversal skills, with a light monitoring framework to continuously assess impact indicators and to steer improvements whenever required.
- **Network building and knowledge sharing**: Leverage complementary expertise and a dedicated project website to disseminate resources, share best practices, and foster European collaboration.

1.2.5 Logo Narrative

ERNEST draft logo is shown in figure 1.1 in b/w although different colored versions will be implemented. **ERNEST is a quantum cat on a scientific mission:** to bring science to life in the most unexpected places. A playful nod to **Schrödinger's cat**, which embodies the curiosity and discovery spirit at the heart of scientific research. The **ERNEST** acronym stands for **European Researchers' Night Escape Science Team**, a creative format that transforms public engagement into an **interactive scientific adventure**.

Figure 1.1: ERNEST project logo in b/w.

Here, the lab becomes a puzzle room, the scientific method your toolkit, and discovery a shared journey. As a quantum cat, ERNEST lives in the space between curiosity and knowledge — just like the ERNs, which opens the doors of science to everyone, across Europe, for one extraordinary evening. Through **hands-on escape games inspired by scientific research**, ERNEST invites participants to experience what it feels like to think, act, and explore like a scientist. It's not just fun—it's science in action. **Much like Schrödinger's cat, trapped in a box of possibilities, participants step into a space where knowledge is revealed only through interaction.** In these escape rooms, participants don't just solve mysteries—they **do science**, uncovering hidden principles, breaking barriers between the public and the research world. From bustling city festivals to quiet school corridors, ERNEST invites everyone to explore how scientific thinking works: **question by question, lock by lock**. ERNEST is a furry ambassador of the ERN spirit: **science for everyone, everywhere — through play, wonder, and discovery.**
#§CON-MET-CM§# #§COM-PLE-CP§# #§REL-EVA-RE§#

2. Impact #@IMP-ACT-IA@#

2.1 Project's pathways towards impact

The following subsections detail the core elements of the project's impact and dissemination strategies, including the rationale and strategic value of ERNEST initiatives, the defined targets and expected impacts, and the anticipated scale and significance of the project's contribution.

2.1.1 Rationale and strategic value of ERNEST initiatives

The **key aspects of the project** are summarized in the following four points:

1. The Need for a Europe-wide science escape room initiative. Local activities with scientific escape rooms already exist in many countries, but these efforts are often fragmented and reach only limited audiences. A European-level project is needed to scale up these immersive science formats across borders, reaching a larger and more diverse audience while aligning with Horizon Europe priorities. By providing a coordinated framework, the project creates a shared narrative about the role of European research in addressing societal challenges. This

approach is expected to increase citizens' understanding of research, enhance awareness of EU-funded projects and positive attitudes toward science, generating impacts beyond immediate project duration.
2. Challenges and opportunities driving this project. The project responds also to other key challenges, for example lack of interactive, first-person learning experiences that make science approachable and fun, and the urgency to achieve gender equality in science. Escape rooms can embed women scientists as role models within their stories and activities, making gender balance a visible, interactive part of the experience. At the same time, there is a clear opportunity: citizens, especially pupils, increasingly respond to game-based learning formats.
3. Urban revitalization in smaller cities. By setting up escape rooms in small and mid-sized towns across Europe, the project can activate unused or neglected urban spaces and turn them into lively cultural hubs. These venues would attract both residents and visitors, helping boost nearby businesses and local services. They also offer accessible entertainment without requiring travel to major cities. In doing so, the project supports decentralization of culture and contributes to the economic and social renewal of less prominent urban areas.
4. What we can achieve beyond existing efforts. This project uniquely enables **cross-border cooperation** among researchers, educators, and science communicators, producing standardized and reusable escape-room materials that can be adopted across Europe. By creating a network of educators and researchers, the project ensures a sustained multiplier effect, with long-term improvements in science literacy and public engagement with EU research that continue far beyond the project's duration.
5. Why ERNEST consortium is best positioned to deliver impact. Our consortium combines expertise in escape-room design, science communication, education, social sciences and humanities and direct collaboration with EU-funded researchers. Established networks of schools, educators, and institutions allow immediate access to target audiences, while our commitment to openly share and transfer through our methodology ensures long-term sustainability. Unlike standard educational escape rooms, which are often developed by game designers, ERNEST involves researchers themselves as inventors and facilitators. Their direct participation ensures that the activities are grounded in current, cutting-edge research rather than simplified or generic content. Through this approach, the project is expected to create a lasting multiplier effect, enabling other organizations to replicate the format, expanding reach across Europe.

2.1.2 Project targets and expected impacts

The project engages specific groups whose involvement directly contributes to **the uptake and exploitation** of the project's results. The **primary target group** consists of **students**. Through their direct participation in the escape-room sessions and school-based activities, they will experience science in a first-hand and playful manner, facilitating a concrete increase in their understanding of scientific concepts and a rise in their interest in STEM subjects (measured through post-activity feedback and follow-up surveys). The **secondary target group** includes **teachers and educators**. By attending training workshops and co-creation sessions, they will learn how to use and adapt the escape-room methodology in their own classes. The **tertiary target group** is the **general public and families**. Their participation in ERN events will generate greater familiarity with EU-funded research and stronger trust in scientific evidence. In addition, the involvement of EU-funded researchers and research institutions will have a dual effect: (i) it will guarantee the scientific quality and relevance of the content delivered to citizens, and (ii) it will help researchers to improve their own communication skills, increasing their willingness to engage in future outreach initiatives. Finally, **local and regional science centers and policymakers** will facilitate the deployment and scaling-up of the activities. Their participation will produce better coordination of existing science engagement resources and ensure that the project results are embedded into regional and national strategies, which is essential to maximize impact beyond the duration of the project. In addition, **media outlets** (local, national and specialized) will be actively engaged through press releases, interviews and media coverage of the events. Their involvement will generate a multiplier effect, allowing the project's results and messages to reach citizens who do not directly participate in the activities, further increasing public awareness of EU-funded scientific research, improving a widespread understanding of modern scientific facts and applications in everyday life and reinforcing the visibility of the EU as the driving force behind the initiative.

2.2 Measures to maximise impact - Dissemination, exploitation and communication #@COM-DIS-VIS-CDV@#

To maximize the impact of the project, we will implement a comprehensive and audience-targeted strategy encompassing **communication**, **dissemination** and **exploitation** activities. All measures will be conceived to reach and engage citizens, pupils and educators in an accessible and stimulating way. At the start of the project, a strong **visual identity (logo, templates, website, social media channels**) will be established to ensure high visibility across all partner countries. Press releases and media engagement around key milestones will help amplify this outreach to wider audiences. To ensure **dissemination** of the results to relevant education and science communication communities, partners will present project outcomes at **science education conferences and big**

events (e.g. Didacta, EAIE, BETT Show, European Youth Event, INTED, EDULEARN, ICIERI) and participate in **public engagement** events. A set of open and user-friendly **teaching resources** (factsheets, guidelines, video tutorials) will be made freely available and promoted via an open-access online repository to encourage replications. Finally, a dedicated **exploitation and sustainability plan** will be developed. This will include long-term maintenance of the escape-room toolkit and the integration of the developed activities into the regular outreach programs of partner institutions. Contacts with **educational authorities** will be fostered to explore opportunities for inclusion in national STEM strategies. A **light monitoring framework** with qualitative and quantitative indicators will be implemented to continuously assess the effectiveness of the communication and dissemination measures and to steer improvements whenever required.

2.2.1 Plan for the dissemination and exploitation including communication activities

The plan includes both face-to-face and digital activities and mobilizes the full outreach capacity of the partner institutions. The scale and significance of the ERNEST contributions are estimated as follows. Starting from the current **baseline situation**—where similar science escape-room initiatives reach **around 2,000 pupils/year**, involve **fewer than 20 trained teachers** and receive **limited media coverage**—the project is expected to significantly expand the number of citizens who come into direct contact with European research through immersive and engaging formats. Over the duration of the action, it is estimated that approximately **4,000–6,000 pupils** will participate in the escape-room activities and school sessions, while **100–200 teachers** will be trained to reuse the methodology in their own educational environments. Participation in ERN events is expected to reach **8,000–10,000 citizens**, and a total of **100–150 EU-funded researchers** will be involved in the delivery of the activities. We also anticipate that the project will attract coverage from **10–15 regional/national media outlets**, thereby extending the project's visibility well beyond the direct participants. These estimates are based on the number of committed schools and partner institutions, the average number of pupils per class (20–25), the assumption that trained teachers will run the escape-room activity at least once per year, and conservative participation figures drawn from recent **ERN reports** in the participating regions. The consortium coordinates partners from **three States** and will create synergies with national initiatives (e.g. Notte dei Ricercatori, Fête de la Science, Research Uncovered) and European networks (EUSEA, Scientix) to widen the dissemination and uptake of results. It is important to emphasize that **all figures refer exclusively to this project** and do not include potential contributions by others.

2.2.2 Plan for EU promotions

All promotional materials will explicitly reference ERNEST as part of a **Europe-wide initiative funded by the European Union**, and the common visual identity provided by the European Commission will be systematically applied. During the events, information panels and short introductory videos will present concrete examples of how EU-funded research contributes to citizens' daily life, including health, environment, and digital technologies. Researchers will explicitly introduce themselves as beneficiaries of MSCA and Horizon Europe projects, empowering the visibility of the EU's role in supporting science and innovation. To reinforce the **ERNEST leading thread of investigative engagement**, a dedicated post-escape-room activity titled **"EU Explorer Challenge"** will be organized. In small groups, participants will solve interactive **EU-themed trivia**, discovering real Horizon Europe and MSCA-funded projects and their impact on everyday life. This debriefing activity is designed to complement the escape-room experience while highlighting the EU dimension in a way that is fully aligned with ERNEST's focus on exploration and hands-on learning. Dedicated promotion of the "Researchers at Schools" activities will continue through direct contact with educational authorities and school networks, targeted newsletters, and dedicated online content. Invitation letters, brochures, and a short video aimed at teachers and headmasters will highlight the educational value of both the escape-room and the "EU Explorer Challenge," ensuring broad participation from schools and pedagogical centers. #§COM-DIS-VIS-CDV§# #§IMP-ACT-IA§#

3. Quality and efficiency of the implementation #@QUA-LIT-QL@# #@WRK-PLA-WP@#

3.1 Work plan and resources

3.1.1 Overall structure of the work plan

ERNEST is organized in **5 interconnected Work Packages (WPs)**:

WP1 – Awareness Campaign: Establishes the communication and engagement backbone. It creates a strong visual identity, maintains digital presence through the consortium website, trains facilitators, engages media and stakeholders, organizes flagship events, and ensures sustainability. WP1 integrates with all other WPs by providing promotion, trained staff, and outreach coordination and drives audience to WP2 and 3 activities.

WP2 – Activities during the NIGHT: Acts as the creative and operational heart. It develops and delivers escape room scenarios for the ERN across multiple cities, both in-person and online. It coordinates venue selection, event

logistics, and reporting. It interacts closely with WP1 for promotion, WP3 for school outreach links and shared material, WP4 for evaluation, and WP5 for management.

WP3 – Researchers at Schools: Brings ERNEST into classrooms, empowering teachers and students through training, co-creation of scientific escape rooms, science investigations, and a puzzle design contest for students. It directly links WP1 for awareness and WP2 for feeding participants and content into ERN events, while WP4 evaluates its impact.

WP4 – Impact Assessment: Defines KPIs and a shared Theory of Change, collects and analyses data, and provides interim and final evaluations. It ensures continuous feedback loops across WPs 1–3, benchmarking against other projects, and guarantees compliance with GDPR and ethical standards.

WP5 – Management: Provides the governance framework of the previous four WPs, with three levels — the Project Coordination Team (day-to-day operations), the Work Package Leaders Group (quarterly technical and strategic review), and the Steering Committee (formal decision-making with partner representation, including young researchers). It ensures financial monitoring, schedule and risk management, consortium cohesion, stakeholder interaction, and structured reporting.

Coordination mechanisms include a **shared digital workspace**, a **centralized outreach calendar**, version-controlled repositories, standardized reporting templates, and regular cross-WP checkpoints. **Sustainability is embedded across WPs**: reusable escape room kits, open-access toolkits, trained facilitators, and long-term commitments from schools, museums, and research institutions ensure that ERNEST's activities extend beyond EU funding.

3.1.2 Timing of the different work packages and their components

The Gantt chart below (figure 3.1) outlines the planned activities, illustrating the structure, sequencing, and interdependencies of the project tasks. **Tasks (T), deliverables (D), and milestones (M)** are labeled as TX.Y, DX.Y, and MX.Y, where 'X' indicates the WP and 'Y' denotes their chronological order within that WP. The months of September 2026 and 2027 are highlighted in yellow as a reminder of the ERN events planned.

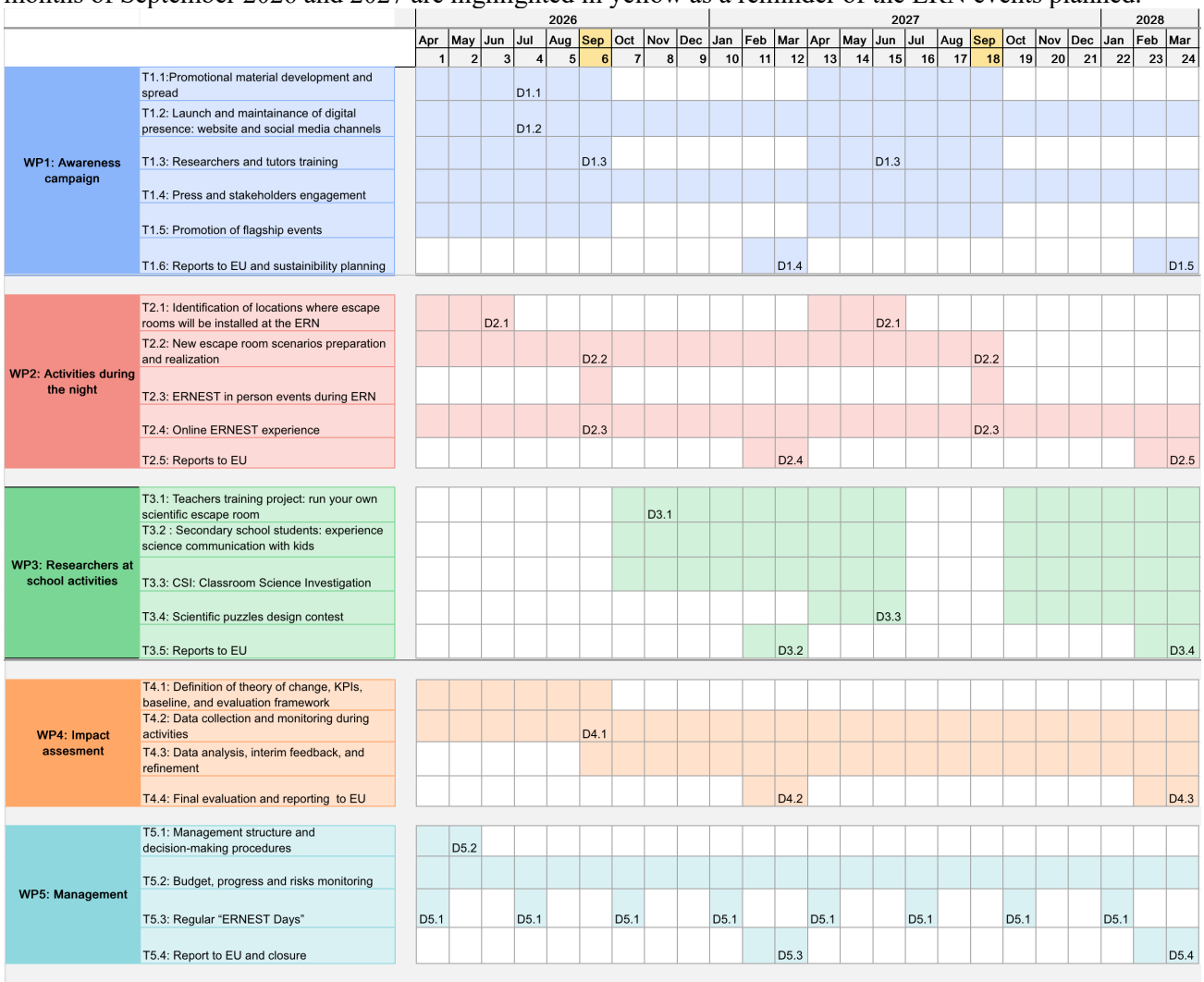

Figure 3.1: Gantt Chart of the ERNEST project.

3.1.3 Detailed work description

This section provides a comprehensive overview of the project's implementation strategy, broken down into structured tables that detail its scientific, operational, and management components. The work is organized into **five WPs (see Table 3.1a)**, each with clearly defined **objectives, tasks, deliverables, milestones, and risk management measures (Tables 3.1b–3.1e)**. The allocation of **staff effort across partners** is summarized in **Table 3.1f,** ensuring a balanced and efficient distribution of resources. The **financial aspects** are detailed in **Table 3.1h**, which cover subcontracting, purchase costs, and other cost categories. Overall, we structured the project on **a total of 100 PM and of these 60 will be in-kind**. The effort is distributed across all partners (WP leaders have a larger weight on their corresponding WP), each contributing a defined in-kind fraction — typically between 50% and 80% of full-time equivalent will be in-kind. Detailed fractions of in-kind contributions per BE are added as comments in the Budget gdoc. This approach ensures full coverage of the planned work while keeping the requested budget within the allowed ceiling.

Table 3.1a: List of work packages

WP No	Work Package Title	Lead Part. No	Lead Part. Short Name	Person-Months	Start Month	End month
1	Awareness campaign	8	UniPg	6	1	24
2	Activities during the NIGHT	1	INFN	11.5	1	24
3	Researchers at Schools activities	11	UniTo	9	7	24
4	Impact assessment	2	INAF	7.5	1	24
5	Management	1	INFN	6	1	24

Table 3.1b: Work package description

Work package number	**1**
Work package title	**Awareness campaign**

Objectives

- Produce engaging and consistent **promotional content** to support network efforts.
- Build and maintain an **active online presence** to increase visibility and audience interaction.
- Equip **researchers and facilitators** with communication skills for effective public engagement.
- Empower **partnerships with media and stakeholders** to broaden project reach.
- Promote **flagship events** that highlight project achievements and attract public interest.
- Monitor progress, **report outcomes, and plan for long-term** continuation of awareness activities.

Introduction to the working package

WP1 is essential to ensure that the innovative ERNEST activities are not isolated events, but part of a coherent, high-visibility, and lasting communication strategy. By combining targeted **promotional content (Task 1.1 or T1.1)** and an engaging **digital presence (T1.2)**, this WP aims to maximize public visibility and community engagement. The **training of researchers and facilitators (T1.3)** ensures consistent, high-quality outreach, while **strategic partnerships (T1.4) and flagship events (T1.5)** provide opportunities to showcase the project's impact. Finally, a **sustainability plan (T1.6)** guarantees the continuation of activities beyond the project's lifetime.

Tasks Breakdown

T1.1: Promotional material development and spread - We'll create a strong, instantly recognizable visual identity for ERNEST that follows the EU's recommendations for funded projects. Our logo will be more than just a graphic: it has a story (described in sec. 1.2.5) and visual elements that link science, play, and discovery. To make the project visible everywhere, we'll produce a **ready-to-use set of templates for posters, banners, slides, and certificates** — all carrying the logo and consistent color scheme. For our team and volunteers, we'll design project **T-shirts, lanyards, and pins**, making them easily identifiable. They **(D1.1)** will be distributed to all partners.

T1.2: Launch and maintenance of digital presence: website and social media channels - Our **website** will be the beating heart of the project **(D1.2)**. It will be a **meeting point for researchers, teachers, students, and curious members** of the public. Visitors will find a) A rich **library of resources**: downloadable templates for escape room puzzles, digital games, educational materials, and step-by-step guides. b) Inspiring content: **videos, photos, and behind-the-scenes stories** from our escape room builds and events. c) **Community space**: a section where teachers can share classroom adaptations, researchers can exchange ideas for new puzzles, and players can

swap tips. d) **Interactive challenges**: online contests where users can collect virtual rewards.
Each participant will have a **personal profile with badges**, pins, and an **"ERNEST passport"** that gets stamped for each thematic escape room or challenge they complete — whether in person or online. Over time, this will encourage repeated participation and create a sense of belonging to the ERNEST community. **Social media** will be used as the conversation space: ERNEST's Facebook, Instagram, X, TikTok and LinkedIn pages will be updated regularly with **mini-challenges, teaser puzzles, event countdowns, and interviews** with scientists and puzzle designers. T1.1 and T1.2 set the stage for an efficient and prompt start of the project **(M1.1)**. Note that given the constrained timeline of the project, T1.1 and T1.2 should start as soon as possible to ensure all the information and material reach the schools before the end of the semester (usually beginning of June).

T1.3: Researchers and tutors training - The training will ensure that all facilitators have the necessary **scientific knowledge, game management skills, and audience engagement** techniques to run the activities in a professional and engaging way. **Training workshops** (in-person and online) will be organized covering: scientific topics simplified explanations for different age groups; best practices in science communication; guidelines for managing small groups during gameplay, ensuring inclusivity and equal participation; safety procedures and handling of equipment or props. In this context we also foresee the involvement of **external experts**: professionals in science communication, public speaking, and escape room designers will be invited to deliver dedicated sessions. These contributions will allow researchers to develop transversal skills beyond their scientific expertise. A **Facilitator handbook** with instructions, troubleshooting tips, suggested discussion prompts, and visual aids will be provided to facilitators, together with **practical mock escape room sessions** where trainees play both participant and facilitator roles, allowing them to test strategies and receive feedback. By forming a new generation of facilitators **(D1.3)** experts in a non-conventional approach to science communication, T1.3 will lay the ground for ERNEST recognition as an **example of innovative science communication in EU (M1.2)**.

T1.4: Press and stakeholders engagement - This task will broaden ERNEST's reach from local communities to international decision makers by building strong relationships with press, media outlets, and key stakeholders in science education and outreach. Key actions include:
Press kit development with press releases, fact sheets, visuals, and videos, aligned with EU guidelines and ERNEST identity. b) **Media outreach** through briefings, interviews, previews, and coordination with partner institutions. c) T**argeted engagement activities** such as press days, open rehearsals, sessions for policymakers, sponsor events, and participation in science festivals and fairs. d) **Media monitoring and impact tracking** to measure coverage, social media reach, and stakeholder feedback, refining the communication strategy.

T1.5: Promotion of ERNEST flagship events - Planned activities of the promotion phase include: a) Training researchers and tutors to act as facilitators and "in-character" guides, enhancing the immersive nature of the events. b) Promoting the events through targeted campaigns on social media, press outreach, and collaboration with schools, museums, and science centers. c) Documenting the events with multimedia coverage and collecting feedback for later evaluation. These events will act as flagship moments for the awareness campaign, building excitement and demonstrating the innovative power of science communication through gamified learning.

T1.6: Reports to EU and sustainability planning - This task will ensure transparent reporting of all WP1 activities and prepare a sustainability framework for the network beyond the project's duration **(D1.4 and D1.5)**. Reporting will follow MSCA CITIZEN guidelines and will integrate data from all tasks, with particular focus on event participation, audience diversity, digital reach, and qualitative feedback on the escape room experience.
Sustainability planning will focus on maintaining and expanding the network after EU funding ends **(M1.3)**. Key actions will include: a) Creating a **sustainability toolkit with reusable escape room** scripts, puzzle designs, and facilitation guides. b) Establishing a **repository of promotional materials, training resources, and event playbooks.** c) Securing **commitments from partner institutions**, science centers, and educational bodies to continue running and adapting the escape rooms. d) Exploring business models, sponsorship opportunities, and integration into existing public engagement programs to ensure **long-term viability**. e) Maintaining the **digital community** — keeping the online platform and badge/passport system active to encourage ongoing participation.

Role of participants and workload quantification

The work of WP1 will be led by the **WP1 Leader (UniPg)**, a partner institution with proven expertise in science communication, event organization, and media relations, supported by all consortium members. Each partner will contribute according to their strengths: content creation from research teams (**CNR, INAF**), digital engagement from social media specialists (**LAMA, UniMan**), training delivery from experienced educators (**INFN/UniTo**). T1.1 is front-loaded in the first 4 months to establish ERNEST's visual identity and produce the promotional

toolkit. Moderate effort will continue throughout the project for updates, printing, and distribution linked to events and new content. Estimated workload: 10% of total WP1 person-months (PM). T1.2 has a high-intensity phase during months 1–4 for website and social media launch, followed by consistent, lower-intensity updates, moderation, and content creation over the entire duration. Estimated workload: 20% of WP1 PM, evenly spread. T1.3 is concentrated in months 1–6 with preparation, delivery of training sessions, and creation of the facilitator handbook. Follow-up and refresher sessions (months 13-18) will occur before major flagship events. Estimated workload: 15% of WP1 PM. T1.4 sees an initial effort in months 1–4 for stakeholder mapping and press kit preparation, followed by recurring engagement activities aligned with event promotion cycles. Estimated workload: 15% of WP1 PM. T1.5 peaks in months 4-6 and 16-18, corresponding to the two main flagship events. Preparation will start 4–5 months before each event, with follow-up evaluations afterward. Estimated workload: 30% of WP1 PM. T1.6 foresees an ongoing data collection and analysis throughout the project, with reporting peaks in months 12 and 24 and sustainability plan finalization in months 20–24. Estimated workload: 10% of WP1 PM. The distribution ensures that creative production (T1.1–T1.3), outreach (T1.4–T1.5), and evaluation/sustainability (T1.6) are interlinked, with overlapping periods allowing for feedback loops between activities.

Coordination and Integration with other WPs
The WP1 Leader will oversee the **planning, implementation, and quality control** of all tasks, acting as the main interface with the Project Coordinator and the leaders of other work packages. **Periodic coordination meetings** will ensure alignment with: a) WP2 and WP3 to synchronize promotional material and training content with the latest game designs and research themes. b) WP4 to share attendance data, participant feedback, and engagement analytics for continuous improvement. c)WP5 to ensure that all reporting and EU visibility requirements are met.

Work package number	**2**
Work package title	**Activities during the NIGHT**

Objectives

- Identify **ERNEST escape room v**enues for both years, securing agreements and safety compliance.
- Design and finalize **new escape room scenarios**, including scripts, puzzles, and setup plans.
- Deliver **in-person and online ERNEST challenges** during the ERNs, including EU promotion.
- Gather **participation and engagement data** to evaluate impact and guide improvements.

Introduction to the working package
WP2 is the creative and **operational heart of the proposal**, and the element that truly sets this project apart from other ERN initiatives. Its **immersive, gamified format** is deployed as a distributed **"spread event" across multiple locations**, including areas that standard ERN activities rarely reach. This approach delivers inclusivity by engaging communities often left out of large urban events, while also posing the main organizational challenge — coordinating high-quality experiences across diverse venues. WP2 builds on our **proven track record of delivering escape room activities at past ERN** editions, where public response has been overwhelmingly positive. This experience provides a strong foundation for developing ERNEST into a **fully structured, dedicated escape room program**, allowing creative ideas and expertise to be refined and exploited to their full potential. Activities will include **securing suitable venues in collaboration with local partners (T2.1)**, designing and testing **new science-themed escape room scenarios (T2.2)**, managing the full **delivery of in person events (T2.3) and online experiences (T2.4)**, and collecting feedback from participants **(T2.5).**

Tasks Breakdown
T2.1: Identification of locations where escape rooms will be installed during the ERN - The selection of suitable venues is central to the success of WP2, as it directly affects accessibility, participant **ex**perience, and the inclusivity of ERNEST activities. Venues will be chosen according to the following specifications: **accessibility, visibility, capacity & layout, technical requirements, safety & compliance, inclusivity and sustainability**. Examples of suitable locations are **public libraries, science centers and museums, community centers in underserved neighborhoods, university campuses, open-air public squares.** These examples demonstrate the adaptability of our portable escape room kits, which can be quickly assembled in a variety of spaces without compromising visitor experience or logistical efficiency. The collaborative approach we plan to follow to identify the best locations of ERNEST events is structured in 2 phases described below.

T2.1.1: Phase 1 - Venue identification, mapping, and partner engagement - Within the first three months of the project, potential venues across all target locations are mapped, prioritizing the ERNEST's thematic and technical

needs **(D2.1)**. Coordination with other ERN projects avoids overlap in the same cities or neighborhoods. Local liaison officers within partner organizations lead venue negotiations and community coordination. An **existing network of trusted local partners** enables swift, reliable venue selection for the first edition. For the second edition, alternative locations will be selected to maximize inclusivity, reaching new audiences and expanding geographic and social coverage.

T2.1.2: Phase 2 - Formal agreements, compliance, and backup planning - Venue agreements, including terms of use, schedules, technical provisions, and safety requirements, are signed a few months before events. A standardized **venue safety & accessibility checklist** ensures compliance with fire safety, emergency access, insurance, and inclusivity standards. To mitigate last-minute cancellations, alternative venues are identified in each city/region as fallbacks. Monthly reports from local liaison officers track readiness and resolve issues promptly. Site visits verify suitability before final approval **(M2.1)**.

T2.2: New escape room scenarios preparation and realization - The consortium enters T2.2 with a rich portfolio of experience in creating and delivering scientific escape rooms that have already proven their capacity to attract, engage, and inspire diverse audiences, described in detail in section 1.2.4.2. Together, these initiatives have reached tens of thousands of participants. The defining strength of these escape rooms is their **adaptability**. Most are designed as **portable, modular kits** that can be installed in a wide variety of spaces, adjusted in length and difficulty, and tailored almost instantly to the age of the audience. Beyond their immediate outreach impact, these **existing formats represent a valuable creative and technical resource for the development of new scenarios**. The setups, props, puzzle mechanisms, and digital tools developed for previous escape rooms can be reconfigured, updated with new scientific content, and combined in innovative ways, reducing production time and costs while ensuring high quality. This reuse also reflects a commitment to sustainability, minimizing waste and maximizing the lifecycle of materials. T2.2 will evolve past successes by introducing fresh narratives, integrating emerging research themes, and experimenting with new formats that expand the reach and impact of ERNEST's scientific escape rooms (see section 1.2.4.3 for our priority list of ideas for novel scientific escape room scenarios).

T2.2.1: Concept development & thematic alignment - Selection of topics in consultation with ERNEST's scientific team, local partners, and other ERN projects to ensure variety, thematic coherence, and alignment with EU science priorities. In developing new scenarios, the consortium will also consider **ad-hoc** scientific themes inspired by the **most relevant and debated research topics of the moment**, ensuring a strong connection with the European scientific landscape. Each escape room weaves together **challenges from across science**—physics, chemistry, biology, mathematics, engineering—and elements of history, geography, or art. This **mirrors real scientific inquiry**, which often spans multiple fields and invites participants with different strengths.

T2.2.2: Co-creation & inclusivity - We will craft each escape room's storytelling through an **iterative design process**—starting from a strong central theme, building a narrative arc with engaging characters and challenges, and ensuring every puzzle fits seamlessly into the plot. To refine the concept, we will conduct a **SWOT analysis** (Strengths, Weaknesses, Opportunities, Threats) to identify what works and what could be improved. All materials will be developed with **multi-language capability** and accessibility features (e.g., large print, accessible fonts, simplified language options, alternative formats). This adaptability has been already demonstrated with HEPscape!, which was translated in multiple languages—including English, French, and German- with SMARTHEP network.

T2.2.3: Technical design & production - We will design **modular escape room kits** that can be easily transported and assembled in a variety of venues identified in T2.1. The puzzles will combine **physical props with digital enhancements**—such as QR codes or augmented reality—to increase interactivity and allow flexible setup. All components will be built from sustainable, reusable materials and will incorporate energy-efficient technologies where possible. Gameplay will be refined through **internal testing** with partner teams to adjust mechanics, difficulty, and timing. We will then run **pilot sessions** with small public groups to gather feedback on clarity, engagement, and accessibility.

T2.2.4: Deployment & knowledge sharing - In the first year, we will focus on perfecting and replicating existing kits while introducing **1–2 new scenarios**, whereas in the second year we will concentrate exclusively on brand-new escape rooms. Each scenario will be completed **at least one month before** its scheduled event. These materials, along with all designs, will be shared with ERN partners to encourage replication and adaptation across Europe **(D2.2)**. Scenarios will **debut during ERN events**, facilitated by **trained staff** to ensure smooth operation and participant engagement **(M2.2)**. **Feedback** will be systematically collected during and after events, feeding into

continuous improvement cycles. The developed resources will be made available to project partners, other ERN organizers, and the wider science engagement community, enabling replication, adaptation, and localization in different European contexts. The trained facilitators, drawn from local communities and research institutions, will **retain the skills and experience gained**, strengthening local capacity for interactive science engagement. The **online ERNEST platform will remain active** after the project, serving as both a repository of past scenarios and a hub for ongoing community interaction, allowing schools, museums, and science centers to access the content and keep audiences engaged all year round.

T2.3: ERNEST in-person events during ERN - The ERNEST escape rooms will be a flagship public activity during the ERN in all participating cities. Participants will be immersed in **richly themed environments** where atmospheric lighting, sound effects, and carefully designed props transport them into the heart of the story.

T2.3.1: Events logistics - We will host these activities in **high-visibility venues**. The physical space will be divided into: a) **Welcome & briefing area**, where visitors receive the storyline introduction and safety instructions. b) **Escape room space**, designed for optimal flow, visibility, and atmosphere. **Puzzles** will be thoughtfully layered, starting with approachable challenges that build confidence before progressing to more complex, interdisciplinary tasks requiring **teamwork and diverse skills.** A visible **countdown timer** will inject a sense of urgency and excitement, driving focus and collaboration. c) **Debrief & interaction zone**, where participants discuss their solutions with researchers and explore the underlying scientific concepts through short demonstrations or displays. To reinforce the European dimension and align with the ERNEST leading thread, a post-escape-room debriefing activity titled **"EU Explorer Challenge"** (described in **2.2.2**) will be offered. To maximize reach and manage audience flow, we will run **multiple sessions per evening** with defined time slots, using a mix of pre-booking and on-the-spot registration. While waiting, visitors will be invited to join short complementary activities—such as hands-on experiments, quick science quizzes, or photo corners linked to the escape room theme—to keep engagement high. Dedicated sessions will be **tailored to specific age groups**, ensuring puzzles and explanations are age-appropriate and engaging. Feedback will be collected in real time via short **surveys, interactive feedback** walls, and informal conversations. Online questionnaires will be provided to the visitors, as detailed in WP4. To create lasting memories and reinforce the unique nature of the experience, all participants receive **ERNEST-branded gadgets** as tangible connection to the scientific journey they undertook.

T2.3.2: Inter-City connections and collaborative challenge - Inspired by the spirit of friendly rivalry and community across participating cities, we will implement a **gamified competition linked to the ERNEST escape rooms**, inspired by the Schrödinger Cat narrative's themes of observation and discovery. Rather than focusing only on speed, the competition will reward cities based on two key metrics: the **total number of visitors** participating in the escape rooms, encouraging broad public engagement; and the **performance on a debriefing quiz**, designed to assess participants' understanding of the scientific concepts explored. Using a Telegram bot accessed via QR codes, players can submit their quiz answers and contribute to **real-time leaderboards that rank cities** by both participation and scientific comprehension. To further enhance the sense of **shared experience across Europe**, selected sessions will feature **live connections between escape rooms in different cities**, where participants tackle special **riddles and quizzes together in real time**. All players will be awarded a personalized **Scientific Passport** stamped with the fields of science explored during their experience, serving as a memento of their journey **(D2.3)**.

T2.4: Online ERNEST experience - In preparation for and during the ERN events, the ERNEST project will establish a **vibrant online platform** designed to gather and engage a community of science enthusiasts across Europe. Each visitor will be able to create a **personal profile**, which serves as a digital hub to track their journey through the ERNEST activities. This profile will showcase their individual achievements, including a **digital version of their Scientific Passport** stamped with badges earned from completed puzzles and challenges. To enhance accessibility and promote participation, the platform will feature an **interactive map** enabling users to easily locate the nearest ERNEST escape room venues in their city or region. Moreover, the website will host a range of **online quizzes and mini challenges** that complement the escape room content, allowing users who cannot attend in person to engage with the scientific themes remotely. These quizzes will contribute to the accumulation of badges and points on their profile, further motivating repeated visits and sustained interest. Additional features may include **forums** or discussion boards for participants to share strategies, insights, and feedback, as well as regular updates highlighting upcoming events, new scenarios, and leaderboard standings.

T2.5: Reports to EU - Throughout the project, ERNEST will ensure documentation and evaluation of all activities conducted during the ERN events. The first year's report **(D2.4)** will focus on in-person events, detailing

attendance figures, participant feedback, and media coverage to capture the public's engagement and the project's immediate impact. Building on this, the second year's report **(D2.5)** will highlight also the online sessions, summarizing participant reach, engagement metrics, and qualitative feedback to assess the effectiveness of the project's digital expansion. The completion of ERNEST events across both years will be marked by a milestone confirming the collection and analysis of comprehensive data, including participation statistics and feedback summaries. These reports and the milestone **(M2.3)** together provide a transparent and structured account of ERNEST's progress, ensuring clear communication of outcomes to the EU and supporting future improvements.

Role of participants and workload quantification

INFN will coordinate WP2, thanks to the expertise in scientific outreach and multi-site event management to oversee venue selection, scenario development, event delivery, and impact evaluation. Scenario development will be shared among partners to reflect their scientific expertise and outreach priorities. Partners will bring complementary strengths and local reach: regional teams will secure inclusive venues not only in their own cities but also in nearby locations where they have established contacts and experience, broadening ERNEST's geographical impact. **All partners** will actively engage in testing, refinement, and data collection, fostering a collaborative feedback cycle that continuously enhances the project. The workload for WP2 is distributed across partners according to their roles and expertise, ensuring balanced and efficient contributions throughout the project. The lead partner will dedicate approximately 20% of total PM to overall coordination, scenario oversight, and quality control. Regional partners will allocate 50% of PM collectively for venue scouting, local liaison, and event facilitation, leveraging their existing networks for broad geographic coverage. Scientific contributors will contribute 20% PM focused on content development and scenario accuracy, while communication and design specialists will invest 10% PM in narrative crafting, puzzle design, and usability testing.

Coordination and Integration with other WPs

WP2 acts as both a content creation hub and an operational driver, requiring tight collaboration across scientific, logistical, and communication teams. Coordination within WP2 includes: a) **Lead Partner role:** The WP2 coordinator will oversee all tasks (T2.1–T2.5), ensuring that timelines, quality standards, and inclusivity objectives are met. The coordinator will chair periodically WP2 meetings to review progress, identify risks, and reallocate resources as needed. b) **Local liaison officers:** Designated by regional partners, these officers will be the primary contact points for venue selection, local stakeholder engagement, and event delivery. c) **Scenario development working groups:** Comprising scientific experts, game designers, and communication specialists from multiple partners, this group will meet periodically during T2.2 to ensure puzzles, narratives, and accessibility features are integrated coherently. Integration with other WPs builds upon the following items: a) WP2 benefits from WP1's outreach to drive attendance, with pre-event interest data helping refine venue selection, scheduling, and accessibility planning. b) School outreach in WP3 generates early engagement for WP2 events, and in some cases acts as a testing ground for new escape room riddles. WP2 provides WP3 with adapted puzzles, narratives, and science content that link school visits to the ERN activities. c) Evaluation methods from WP4 are embedded into WP2's in-person and online experiences from the outset. WP2 supplies WP4 with participation statistics, engagement metrics, and qualitative feedback to measure impact and guide improvements for future editions. d) WP5 supports WP2 with contractual, scheduling, and risk management for venue agreements and event delivery. WP2 delivers progress updates, risk alerts, and budget usage data to WP5, ensuring smooth coordination and compliance with project timelines.

Work package number	**3**
Work package title	**Researchers at schools activities**

Objectives

- Empower **teachers to integrate interactive science engagement into curricula** by providing training, resources, and hands-on experience to design and run their own scientific escape rooms.
- **Enhance students' science communication and problem-solving skills** through direct collaboration with researchers in structured non-conventional activities.
- Encourage **creativity, teamwork, and scientific literacy by organizing a Europe-wide scientific puzzles** design contest, encouraging students to create original, research-inspired challenges.

Introduction to the working package

WP3 brings the **ERNEST experience directly into schools**, transforming classrooms into spaces of scientific exploration, creativity, and collaboration. Teachers will take part in a **dedicated training program**, **Run Your**

Own Scientific Escape Room, gaining the tools, resources, and confidence to deliver their own science-based escape experiences **(T3.1)**. WP3 engages high school students, giving them **first-hand experience in science communication** by co-delivering activities for younger audiences **(T3.2)**. Finally, WP3 turns classrooms into investigative labs **(T3.3)** with **CSI: Classroom Science Investigation**, where middle-students tackle hands-on, inquiry-driven challenges guided by real researchers. To encourage creativity and teamwork, in the second year of WP3, we plan to launch a **Scientific Puzzles Design Contest (T3.4)**, inviting students across participating schools to design original, research-inspired puzzles, with recognition for innovation and problem-solving skills. Finally, we will ensure thorough documentation, impact assessment, and **reporting to the EU**, consolidating results from all activities and identifying opportunities for improvement and long-term adoption **(T3.5)**. **Continuous feedback** from teachers, students, and partners will drive iterative improvement, ensuring that WP3 not only delivers engaging experiences but also leaves a lasting educational legacy. To ensure that students are exposed to diverse role models, ERNEST will guarantee **gender-balanced participation of researchers in school-based activities**. At least half of the scientists will be women, representing a wide spectrum of disciplines and career stages.

Tasks Breakdown

T3.1: Teachers training project: run your own scientific escape room - Imagine walking into a classroom where students aren't just learning science — they're living it. The lights dim, a lock clicks shut, and a story begins to unfold. To "escape", students must solve scientific riddles, decode data, and think critically together. In this teacher training project, we give educators the tools, resources, and confidence to bring this kind of immersive learning to life — right in their own schools. This initiative transforms teachers into **game designers of science**, guiding them through the process of creating their own scientific escape rooms. It is built on four phases:

T3.1.1: Kick-off workshop: from players to learners - Teachers gather for an interactive first session with researchers where they play one or two complete scientific escape room activities. The **workshop is fully hands-on**: participants experience the puzzles, narrative, and time pressure exactly as students would. Afterwards, we break down the **game mechanics**: how clues are built, how puzzles connect to science topics, and how storytelling can motivate learning.

T3.1.2: From player to creator: DIY escape room kit - Each teacher receives a **Do-It-Yourself Scientific Escape Room Kit**, which includes: a) Step-by-step creation guidelines and puzzle templates. b) Printable props, story outlines, and scientific challenges linked to school curricula **(D3.1)**. c) **Access to the ERNEST project website** with downloadable games, material and inputs. This stage is about demystifying the creation process — showing that you don't need special effects or big budgets to make an engaging science game.

T3.1.3: Co-creation phase: design in the classroom (1–2 months) - Teachers work with their students to adapt a kit or create an entirely new game under the supervision of one researcher and exploiting: a) **Curriculum alignment,** i.e. games can be built around topics such as chemistry, physics, biology, climate change, or space exploration. b) **Multilingual exchange**: some classes may focus on translating existing games into new languages, creating bridges between schools across countries. c) **Inter-school connection**: teachers and students can share prototypes, get feedback, and even swap puzzles with partner schools in other cities. Classrooms can use **existing science equipment**, recycled materials, and art supplies to create clues and props. Nearby universities or research institutes allow teachers to integrate real scientific data into their challenges. **Colleagues from other subjects**, parents, and even local volunteers can join in as co-creators turning the activity into a truly collective effort.

T3.1.4: Final meeting: showcase and exchange - The last meeting is a celebration and learning exchange. Teachers present their classroom experience, successes, and challenges **(M3.1)**. A feedback session gathers insights from both students and teachers, which will be used to improve the kits and resources for future use.
Expected outcomes include: teachers confident in using **game-based learning for science, a growing, multilingual repository of ready-to-use escape room riddles,** lasting **partnerships** between schools across different cities and countries. By the end of the project, we will have seeded a **sustainable, collaborative network** of teachers and students who can continue to create, adapt, and share scientific escape rooms, inspiring future generations well beyond the project's lifetime. The DIY kits, guidelines, and materials developed during the project will remain available through the ERNEST repository.

T3.2: Secondary school students: experience science communication with kids - What happens when **teenagers become science facilitators**? In this activity, secondary school students (upper secondary, typically aged 15–18) step into the role of **science communicators**, running hands-on activities to spark curiosity in younger children.

This activity offers a structured **work-based learning experience** — known in Italy as "Formazione Scuola-Lavoro". Over several weeks, they will learn to run and deliver interactive science activities for primary or lower secondary school children, gaining hands-on experience in simplifying complex topics, adapting their communication style, and inspiring curiosity. This task builds both **science outreach capacity** and **transferable skills** among older students and includes three phases:

T3.2.1: Training and orientation - Secondary school students participate in an introductory workshop led by researchers and science communication experts. They explore **effective methods for explaining scientific concepts** to children, including storytelling, demonstrations, and game-based learning. They learn first the scientific content of the selected topic, and as an added value they learn how to explain it to younger visitors. They learn about **audience engagement**, clear language, and techniques to stimulate curiosity and active participation from children aged 6–10. Whenever possible, activities integrate **local scientific research** or community-relevant topics (e.g., biodiversity, renewable energy).

T3.2.2: Delivery and interaction - Students deliver their activities in real settings such as primary schools, libraries or open days at research institutions. Our portable escape room kits will be set up directly in school spaces, transforming classrooms or common areas into immersive science challenges. Younger students will be invited to take part in the escape room experience, with **secondary school students acting as facilitators** and "game masters," guiding the activity and ensuring a smooth, engaging experience. **Researchers may join** the sessions to connect the activities to current scientific work, adding authenticity and relevance.

T3.2.3: Reflection and recognition - The activity concludes with a **structured debriefing** where students reflect on: a) What worked well in engaging children with science. b) What challenges they faced and how they adapted. c) How the experience changed their understanding of science communication and research. Participants receive a **certificate of participation** recognizing the skills they developed — public speaking, teamwork, creativity, adaptability, and scientific literacy — which they can include in their educational and career portfolios.

Expected outcomes foresee: Secondary school students trained and experienced in **communicating science to non-expert audiences;** younger students are exposed to **inspiring science activities led by relatable near-peer role models;** increased understanding of the role of research and its relevance to everyday life. This model can be repeated annually and adapted to different contexts, connecting new cohorts of students with researchers and younger learners. Over time, it will create a **pipeline of confident science ambassadors** who will carry their communication skills into further studies, careers, and community engagement.

T3.3: CSI: Classroom Science Investigation - What if the classroom became a crime lab? In this activity, junior high students (ages 11–14) become **junior investigators**, using scientific methods to solve hands-on mysteries. Each investigation revolves around a **science "case" designed to** engage students in **critical thinking, experimentation, and teamwork**, in 4 steps:

T3.3.1: Introduction and case briefing - Students are divided into small investigation teams and introduced to a **fictional mystery**, each with a case file containing clues, initial observations, and background information. Possible cases include: **The Leaky Container, The Floating Coin Mystery, The Disappearing Ink, The Broken Bridge, The Mysterious Sound,** etc. Teachers or researchers introduce the **scientific method** as the framework for investigation: observe, hypothesize, test, analyze, and conclude.

T3.3.2: Investigation phase - Teams carry out experiments and tests to replicate or explain the mystery. Students make observations, record data, and test multiple hypotheses. Activities are safe, reproducible in a classroom, and directly linked to physics, chemistry, and engineering concepts.

T3.3.3: Collaboration and reporting - Each team prepares a **brief report or presentation** summarizing their methods, observations, and conclusions. Students practice **clear science communication**, explaining their reasoning and results to peers.

T3.3.4: Debrief and reflection - Class discussion compares investigative approaches and results. Teachers and researchers highlight the **scientific concepts applied**, reasoning skills used, and relevance of inquiry-based learning. Students reflect on how **investigation and experimentation** help solve real-world problems.

Expected outcomes include: enhanced understanding of scientific methods and key concepts in science; development of critical thinking, problem-solving, teamwork, and communication skills; engaging, story-driven learning that makes science tangible and exciting. Teachers can rotate through different cases or combine them for longer projects, creating a **scalable model for science engagement.**

T3.4: Scientific puzzles design contest – What if students could turn the science they love into puzzles that challenge—and teach—others? The contest gives young learners from different countries the chance to become science creators, **designing original riddles that combine curiosity, logic, and hands-on investigation**. By dividing participants into **two age categories (11–14 and 15–18)**, the contest allows students to focus on science topics most relevant to their level, often inspired by subjects they have just studied and found particularly fascinating. In the **design contests**, ERNEST will appoint 50% female researchers **as mentors** to guide student teams in developing their ideas. These mentors will serve as accessible role models, particularly for young women interested in science and technology. In addition, at least one set of riddles or puzzles will be explicitly themed around **gender equality in science**, using historical and contemporary examples of women scientists whose contributions transformed their fields. All submissions are made as **2–3 minutes video presentations**, a format chosen to make participation simple, engaging, and inclusive. The contest unfolds over the **final six months of the project**: **Month 1 -** Online launch and briefing, with instructions, guidance, and examples shared with participants **(D3.3)**. - **Months 2–4:** Students design their puzzles, create videos, and submit their entries through a secure platform. - **Month 5:** A panel of **science educators, project researchers, and communication experts** evaluates the submissions based on scientific accuracy, clarity, creativity, and engagement. - **Month 6:** Results are announced in an online celebration, highlighting the **top three winning puzzles**. The three winning puzzles will be **implemented and integrated into the project's escape room activities**, becoming part of the hands-on experiences offered to other students. The young designers will be **formally recognized as co-authors of the activities**, giving them ownership and visibility for their creative contribution. This strengthens the link between classroom creativity and real-world science communication. All submissions are made with **parental consent**, personal information is limited to first names and school, and content complies with GDPR and national data protection regulations. All students receive **certificates**, and winning puzzles are featured on the project website. Through this contest, students apply scientific knowledge creatively, develop teamwork and communication skills, and contribute directly to an international network of science learning experiences, while seeing their ideas **come to life in real educational escape rooms**.

T3.5: Reports to EU - This task ensures timely and accurate reporting to the EU. All consortium partners contribute to the preparation of periodic technical and financial reports, including progress updates, deliverable submissions, and milestone achievements **(D3.2 and D3.4)**. In the first year, three out of four main WP3 activities will be implemented, followed by a structured reflection and evaluation phase to identify improvements and lessons learned **(M3.2)**. In the second year, activities will be refined based on this feedback, and the contest will be added **(M3.3)**. The WP leader coordinates the collection of information, verifies consistency with project objectives, and compiles the reports. Clear internal procedures for review and approval are established (jointly eith WP5) to guarantee high-quality submissions and compliance with EU requirements.

Role of participants and workload quantification

The lead partner for WP3 is **UniTo supported by UniUPO**, bringing extensive experience in science education, outreach, and innovative learning methodologies. As coordinator, the **lead partner is responsible for overall planning, monitoring, and reportin**g of all WP3 activities, ensuring milestones are met, and maintaining consistent communication with project partners. However, all partner institutions actively participate in the activities, each implementing **at least one of the sub-tasks in their local context**. While the core expertise and experience for these activities are historically rooted in Italy, the **involvement of international partners** adds significant value. A key role will also be played by **stakeholders** such as local schools and regional institutions, with whom the project team has already established strong collaborative relationships. The workload for WP3 has been carefully structured to ensure balanced participation and clear responsibility across partner institutions. Each sub-task represents a portion of the total WP3 effort, distributed according to planning, preparation, delivery, and reporting requirements: **T3.1: 25%, T3.2: 20%, T3.3: 25%, T3.4 and T3.5: 15% each.**

Coordination and Integration with other WPs

Each sub-task within WP3 has a **designated coordinator** responsible for overseeing planning, implementation, and monitoring: T3.1: **Anna Vignati (leader of UniTo)**, T3.2: **Elisa Corino (UniTo)**, T3.3: **Livia Soffi (INFN),** T3.4:

Marta Ruspa (UniUPO). Coordinators ensure that activities align with WP3 objectives, timelines, and quality standards, while all institutions contribute by implementing at least one activity and supporting others. WP3 is closely **linked to other work packages**: it builds on the awareness campaign developed in WP1, incorporates scientific content, connections, and materials such as new riddles and quizzes into WP2, and contributes to evaluation and sustainability planning in WP4. Moreover, all school students who **participate in WP3 will serve as the first cohort of participants for the ERN events organized within WP2**, helping to promote and disseminate project activities. Their involvement will amplify engagement across schools, fostering high participation and extending the reach of the project's educational and outreach initiatives.

Work package number	**4**
Work package title	**Impact assessment**

Objectives

- Define **quantitative and qualitative** KPIs that link back to WP1-3 specific objectives reflecting both participant-level outcomes (e.g. learning, engagement, motivation) and model-level effects (e.g. replicability, inclusiveness, outreach capacity).
- Capture outcomes at multiple levels: at the **individual level** (students, teachers, researchers - short-term changes in knowledge, curiosity, and self-efficacy toward science); at the **organizational level** (schools, science centers, partners - adoption of innovative, playful, and research-based educational formats); and at the **societal level** (local communities - increased access to science engagement in socially and spatially underrepresented areas).
- **Integrate data collection into activities**, combining light-touch quantitative and qualitative tools (e.g. quick polls, feedback corners, brief post-event surveys) and participatory approaches.
- Ensure all surveys and data storage comply with **GDPR**. For minors, get parental/guardian consent.

Introduction to the working package

WP4 will track both **how many people we reach** and **how deeply we inspire them**, focusing on short-term changes among participants and the overall effectiveness of the project model. We will set clear indicators from the start—such as participation, engagement, learning, and motivation—working with education and data analysis experts to ensure high quality and reliability.

The impact assessment will operate on two complementary levels:

- at the **individual and organizational leve**l, it will capture measurable short-term changes among key stakeholders in terms of scientific understanding, curiosity, and perceived relevance of science.
- at the **model lev**el, it will assess the effectiveness, inclusiveness, and perceived replicability of the ERNEST approach as an innovative science engagement format across different contexts.

Feedback will be collected both **online and in person**, so everyone can contribute easily and securely. All instruments will be embedded into activities and aligned with the Theory of Change and Evaluation Framework.

After the first year, we will publish an **interim analysis** to improve activities for Year 2, comparing our results with similar EU projects. The final evaluation will highlight long-term changes and share recommendations for future initiatives. By combining rigorous analysis with open sharing—through conferences, publications, and public summaries—WP4 will not only measure impact but also help others build on our experience.

Tasks Breakdown

T4.1: Definition of theory of change, KPIs, baseline, and evaluation framework - We will define a set of KPIs, using input from educational researchers and evaluation specialists. The process will start with a Theory of Change (TOC) workshop, engaging partners in co-defining the project's pathways of change, underlying assumptions, and intended outcomes at both participant and system level. This shared mapping will align all indicators with the objectives of WP1–3 and with the broader goals of the MSCA & Citizens program. Baseline values will be collected at the start of the project for comparison over time. Questionnaire design will draw on the consortium's **past outreach and citizen science projects**, ensuring accessibility, and relevance to each target.

The indicators will capture outcomes at two complementary levels: a) **Short-term outcomes** for key stakeholders – students, teachers, and researchers. b) **Effectiveness and replicability** of the engagement model – as perceived by educators, researchers, and consortium partners. Indicators will include:

- **Reach and Participation**: number of students, teachers, researchers, schools, and communities involved.
- **Engagement**: participation rate in activities, time spent in events, completion of interactive tasks.
- **Learning Outcomes**: change in knowledge or skills, self-reported confidence in STEM topics.
- **Motivation and Interest**: increased willingness to engage in future science-related activities.

- **Collaboration**: number and quality of cross-institutional and international partnerships formed.
- **Societal Impact**: perceived relevance of science to everyday life, intention to share science with others, supporting the decentralization of culture, and contribute to the social renewal of less prominent urban areas.
- This task will result in a **shared monitoring framework**, integrating TOC logic, indicators, and data collection protocols **(D4.1)**, to guide the subsequent evaluation activities across all work packages.

T4.2: Data collection and monitoring during activities - Questionnaires and feedback forms will be distributed both o**nline** and **in person** to maximize response rates. Online tools will be used for schools, contests, and post-event follow-up, while **on-site collection** will happen at ERN events and other live activities **(M4.1)**. For in-person settings, we will: a) Provide **tablet stations** or QR codes at the exit of escape rooms and workshops; b) Organize "**feedback corners**" with trained facilitators encouraging visitors to share their thoughts in exchange for small thematic giveaways (e.g., science bookmarks, stickers); c) Integrate micro-feedback moments into the activity flow (e.g., one-question polls before leaving). In addition, LAMA will train facilitators and local staff on data collection protocols, and inclusive engagement techniques. This dual approach ensures that those without immediate online access can still contribute, and that feedback is captured while the experience is fresh.

T4.3: Data analysis, interim feedback, and refinement - All data collected through questionnaires, interviews, and observational tools will be systematically analyzed using a **combination of quantitative and qualitative methods**. Quantitative data (e.g., participation numbers, pre/post knowledge scores, rating scales) will be processed using descriptive statistics, correlation analysis, and, where relevant, comparative tests to identify significant changes relative to baseline values. Qualitative data (e.g., open-ended comments, focus group notes) will undergo thematic coding to capture recurring patterns, unique insights, and unexpected outcomes. We will collaborate with experts in education, science communication, public engagement, and data science/statistics to ensure robust interpretation of the findings. Their role will be twofold: a) **Validation** of the methodology and statistical soundness; b) **Interpretation** of results considering pedagogical and engagement theory. To place our results in a broader context, we will conduct a **benchmarking exercise** comparing our impact indicators with those from similar European initiatives, particularly other MSCA Citizens or ERN projects. This will help to identify strengths, areas for improvement, and opportunities for collaboration or replication. In selected schools, participants will experience a **comparative activity assessment**: both a traditional lesson and thematic escape room on the same subject. Immediately afterward, they will complete the standard questionnaire. This design allows us to directly **compare learning outcomes, engagement levels, and correct-answer rates** between the two formats, providing actionable insights on which method is more effective for knowledge, retention, and motivation. Data from this sub-activity will be integrated with other WP2 and WP3 results for a harmonized analysis. After the first year, we will take stock of progress by producing an **intermediate analysis report** that distills key trends, lessons learned, and actionable recommendations, serving as a strategic foundation for enhancing activities in the second year **(M4.2)**. A mid-term **learning touchpoint** within the consortium will review progress, update the Theory of Change, and ensure consistent advancement toward project impacts. The interim results will not only be used internally but also **shared with the wider professional community**: a) Presentation at **thematic conferences** as for example EUSEA Annual Conference (European Science Engagement Association), PCST Conference (Public Communication of Science and Technology), and relevant EUSEA/ERN impact workshops; b) Publication of short **open-access articles** in journals such as JCOM – Journal of Science Communication or Research for All; *c)* Organization of a **stakeholder roundtable** with educators, outreach coordinators, and institutional representatives to discuss implications and gather feedback.

T4.4: Final evaluation and reporting to the EU - The final **impact assessment reports** will synthesize each year of data **(D4.2, D4.3)**, showing trends in engagement, motivation, and learning. They will highlight the project's role in supporting international collaboration, strengthening STEM culture, and inspiring long-term involvement in science. The reports will include recommendations for replication and scaling, ensuring the **legacy of our impact**. All results will be presented in an engaging format for stakeholders and shared in public summaries and conferences to maximize transparency. This process will culminate in the **comprehensive impact analysis** at the end of the project, with the final evaluation report formally reviewed and endorsed by the network **(M4.3)**.

Role of participants and workload quantification

The lead partner for WP4 will be **INAF**, based on its prior experience in the evaluation of educational resources and public outreach events to coordinate the impact assessment and dissemination of activities. **LAMA will act as the consortium's expert partner in social impact evaluation and educational inequality**, supporting INAF in the design and implementation of the impact assessment methodology. LAMA will lead the development of the

TOC and Evaluation Framework, ensuring methodological consistency across data collection and analysis tools, and providing training to facilitators and partners involved in field monitoring. It will also be responsible for drafting interim and final impact reports, and for coordinating learning touchpoints to review progress toward project outcomes and impacts. INAF will be responsible and coordinator of the internal dissemination actions, supported in any aspect by other members of the consortium such as **INFN, CNR, UniBa and UniTo**. All consortium partners will actively contribute by: a) **Schools and educational partners:** administering questionnaires, facilitating in-person feedback during events, and supporting baseline data collection; b) **Research institutions:** providing scientific expertise, data analysis support, and review of methodological approaches; c) **International partners:** offering benchmarking comparisons with similar projects abroad, sharing insights, and helping interpret results in diverse educational contexts. The distribution of WP4 efforts among the consortium partners is expressed as a percentage of total WP PM. These percentages are calculated based on the complexity, time intensity, and required coordination for each task: **T4.1 – 20%:** This reflects the preparatory nature of defining TOC, KPIs, baselines, and Evaluation Framework. It involves substantial planning and consultation with experts but requires less continuous effort once tools and indicators are established. **T4.2 – 30%:** Data collection and monitoring is resource-intensive, requiring consistent partner involvement across multiple events. The higher allocation accounts for logistical coordination, participant management, and real-time quality control of data. **T4.3 – 35%:** The largest share reflects the time and expertise needed for comprehensive data analysis, interim reporting, benchmarking, and refinement. Engagement with external experts and iterative validation of findings adds to the workload. **T4.4 – 15%:** Final reporting requires consolidation of all data, drafting of deliverables, and internal review, which is less time-consuming but critical for formal submission and dissemination.

Coordination and Integration with other WPs

WP4 is closely linked to all other work packages, ensuring a coherent and comprehensive assessment of the project's impact. To maximize comparability and integration, all monitoring instruments will be drafted in English to maintain consistency across partners. They will then be translated into the relevant local languages by the consortium, ensuring accessibility while preserving meaning. While WP2 will collect detailed data from ERN events, WP3 activities (classroom sessions, school outreach) will also be monitored using aligned instruments. Collecting data in both WPs allows us to compare engagement, learning, and motivation levels across different activity formats. We will identify which formats are most effective for different targets and we will integrate datasets to provide a unified analysis of the project's overall impact, while accounting for context-specific differences. Standardized KPIs, scoring scales, and survey questions will allow meaningful aggregation of results from WP2 and WP3. This approach ensures that insights are both granular (activity-specific) and holistic (project-wide), supporting evidence-based improvements and reporting. Key indicators will be **consistent across WP2 ERN events and WP3** classroom activities, enabling **comparison and integration of results**. Quantitative data will be adjusted for differences in activities, while qualitative data will be coded to capture specific insights.

Work package number	**5**
Work package title	**Management**

Objectives

- Implement a clear, efficient, and participatory **governance framework**.
- Empower consortium cohesion, **partner engagement, and stakeholder interaction.**
- Provide continuous **monitoring** and **structured reporting**, evaluation, and project closure.

Introduction to the working package

We designed WP5 to combine **proven practices with the specific needs of a multi-partner, citizen-focused project**. This approach gets its strengths from our experience in coordinating complex, interdisciplinary teams, balancing operational agility with strategic oversight. At the core of WP5 is a **three-level management structure described in T5.1**. WP5 also addresses **financial oversight, risk management, and interactions with both stakeholders and other ERN projects**. To support learning, cohesion, and proactive problem-solving, WP5 introduces **regular "ERNEST Days"**—informal consortium review meetings designed for open dialogue, knowledge exchange, and collaborative reflection. Finally, WP5 provides a **structured project closure**, producing comprehensive management reports for the EU that document achievements, lessons learned, and recommendations for future MSCA Citizens initiatives.

Tasks Breakdown

T5.1: Management structure and decision-making procedures - The project will be governed through a clear

and efficient three-level management structure that ensures both agile day-to-day coordination and inclusive strategic oversight. At the core of this structure, the **Project Coordination Team (PCT)** – composed of the Coordinator (**Livia Soffi, INFN**), Project Manager (**Francesca Cavallari, INFN**), and Financial Officer (**Matteo Bauce, INFN**) – guarantees the **continuous operational execution** of the action. It is responsible for administrative management, financial monitoring, reporting preparation, communication with the EU, and for implementing all social and engagement-related actions approved at higher levels (e.g., mentoring of weaker partners, rotation of roles, informal exchanges). The **Work Package Leaders Group (WPLG) (Valentina Mariani, UniPg for WP1, Livia Soffi, INFN for WP2, Anna Vignati, UniTo for WP3, Claudia Mignone, INAF for WP4, Francesca Cavallari, INFN for WP5**) forms the second level and brings together all WP leaders, supported by the PCT. Meeting on a **quarterly basis**, this group provides a dedicated **technical and operational forum**, where progress is reviewed, interdependencies are identified, and social aspects (partner engagement, inclusiveness, balanced participation, and diversity) are regularly monitored. When needed, the WPLG prepares concrete proposals for corrective actions (e.g., redistribution of tasks, additional support for less-represented partners). The **Steering Committee (SC)** constitutes the third level and acts as the **formal decision-making body** of the project. It consists of one representative per partner **plus one Young Researchers' Representative**, whose inclusion reflects the dynamic and participatory nature of MSCA Citizens projects, ensuring that the perspectives and engagement of early-career staff are incorporated into strategic decision-making. The SC meets **regularly** to review the proposals prepared by the WPLG, approve changes, endorse risk-mitigation and social-management measures, and oversee compliance with the project objectives and MSCA Citizens requirements. Once a decision is adopted by the SC, the PCT is responsible for its implementation. **Information flows upwards** (from PCT to WPLG to SC), and **decisions flow downwards**, ensuring full internal cohesion and partner ownership throughout the project. **All decisions follow a structured procedure formally approved (M5.1)**, with proposals discussed and agreed in the WPLG before being formally approved by the SC, ensuring transparency, accountability, and timely implementation across all work packages **(D5.2)**. The PCT will also coordinate all **interactions with external stakeholders**, including policy makers, educational institutions, and local communities. A structured **stakeholder engagement plan** will be developed, assigning specific partners and WP leaders to maintain regular contact, collect feedback, and ensure alignment with project goals. Input from stakeholders will be systematically communicated to the WPLG and SC to inform decision-making and strategic adjustments. The PCT will finally coordinate strategic interaction with other ERN projects, ensuring alignment, synergy, and knowledge exchange. Specific WP leaders from different geographical sites will be assigned as ERN liaisons to maintain regular communication with parallel initiatives, share updates, and identify opportunities for collaboration or joint events. Recommendations and lessons from these interactions will flow upwards to the WPLG and SC, enabling informed decision-making and integration into project planning. **In addition, ERNEST will be included and regularly updated on the national portal www.nottedeiricercatori.it** further strengthening its visibility and connection with the broader Italian Researchers' Night community. Similar visibility actions will be pursued in France and the United Kingdom, ensuring that ERNEST is integrated into national communication hubs across partner countries and reaches a wide European audience.

T5.2: Budget, progress and risks monitoring - Ensures the **financial, operational, and strategic health** of the project throughout its entire duration. The **PCT** is responsible for continuous **budget tracking**, verifying that expenditures align with planned allocations, and flagging any potential deviations early. Monthly financial reviews will be conducted, complemented by **quarterly internal audits** to guarantee accurate reporting and transparency. Progress monitoring is implemented through **regular collection of KPI metrics, milestone achievements, and deliverable status updates** across all work packages. The **WPLG** consolidates this information and prepares a comprehensive **quarterly progress report**, which is then submitted to the **SC** for review and decision-making. This reporting cycle ensures that **strategic corrective actions** can be adopted promptly, avoiding delays and maintaining project coherence. **Risk management** is fully integrated into T5.2. The PCT maintains a **live risk register**, updated continuously with input from WP leaders. Risks are classified according to **likelihood and impact**, and mitigation measures are tracked and evaluated during each WPLG meeting. The SC formally reviews high-level risks at each monthly meeting, approving new mitigation measures or resource reallocations when necessary. Special attention is given to **partner engagement, financial contingencies, public health restrictions, and the innovative escape-room activities**, with proactive measures designed to maintain both **project continuity and impact**. **Information flow:** The **PCT** continuously monitors and collects data on budget, progress, and risks. The **WPLG** consolidates and analyzes these inputs, preparing recommendations. The **SC** reviews and approves high-level decisions, ensuring that corrective measures and resource adjustments are **transparent, timely, and strategically aligned** with the project's objectives. All monitoring activities are **documented systematically**, forming the basis of the annual **management reports** and informing decisions on budget reallocations, corrective

actions, and future planning. Interactions with other ERN projects will be monitored as part of **risk management and progress tracking**, identifying potential overlaps, redundancies, or opportunities for cooperation. Any strategic adjustments, such as reallocation of resources for joint activities, will be approved by the SC to maintain project coherence and maximize EU impact.

T5.3: Regular "ERNEST Days" - Establishes a series of informal consortium review meetings that imply **collaboration, knowledge exchange, and proactive problem-solving**. These events provide a dedicated space for partners to discuss ongoing progress, share experiences from their respective work packages, and identify any **operational or social challenges** before they escalate. The **PCT** organizes and facilitates each session, preparing an **agenda, key discussion points, and relevant progress reports**. The **WPLG** actively participates, presenting updates on deliverables, milestones, and risk management. The sessions are structured to encourage **open dialogue**, with particular attention to **partner engagement, inclusiveness, and the integration of early-career perspectives**, including input from a **Young Researchers' Representative**. ERNEST Days will include dedicated sessions to **review ongoing stakeholder interactions**, share lessons learned, and discuss strategies to enhance outreach and collaboration. Each "ERNEST Day" results in **summary notes (D5.1)** that document key discussions, decisions, and action points. These notes serve as an **informal decision support tool** for the SC and help maintain **transparency and accountability** throughout the consortium.

T5.4: Report to EU and closure - The **PCT** leads the preparation of the final management report, consolidating input from all work packages, and ensuring that **budget execution, deliverable completion, and risk mitigation actions** are fully documented. The **WPLG** contributes detailed updates on each WP, highlighting **milestone achievements, KPIs, dissemination activities, and social engagement outcomes**, including feedback from the **Young Researchers' Representative** on participation and inclusiveness. This structured input guarantees that the final report reflects both **quantitative progress and qualitative insights**, providing a rich basis for evaluation. The **SC** reviews and formally endorses the closure report, validating the **accuracy, completeness, and strategic relevance** of the documented outcomes. Additionally, the SC oversees the approval of **lessons learned**, identifying best practices, challenges, and recommendations for future MSCA Citizens projects. A **mid-term assessment (M5.2)** will provide an updated overview of the project's status, including progress, budget consumption, and risk management, allowing timely adjustments and informed decision-making. The intermediate and final management reports **(D5.3 and D5.4)** will summarize **stakeholder engagement activities, outcomes, and lessons learned**. Recommendations for future MSCA Citizens projects will include best practices for maintaining strong, structured stakeholder interaction. All outputs from T5.4 serve as **formal deliverables for the European Commission**, ensuring transparency, accountability, and compliance with Horizon Europe standards. T5.4 **closes the project in a structured and fully accountable manner**, leaving a clear record of achievements, effective governance, and practical recommendations for future initiatives **(M5.3)**.

Role of participants and workload quantification

The Project Coordinator, supported by the PCT, will lead WP5, ensuring overall governance, decision-making, and reporting to the European Commission. In addition to PCT, WPLG and SC, whose roles have been already described in section T5.1 we will have also **All Partners** actively contribute to ERNEST Days, provide inputs for monitoring, and ensure implementation of corrective actions. A key role is played by the **Young Researchers' Representative.** He/She ensures early-career perspectives in decision-making and supports engagement and social cohesion initiatives. WP5 spans the **entire project duration (M1–M24)** and involves the following effort: **PCT:** ~35% of total WP5 effort, distributed across management, monitoring, and reporting tasks; **WPLG:** ~25% of WP5 effort, mainly quarterly progress reviews, KPI consolidation, and recommendations preparation; **SC:** ~25% of WP5 effort, focused on monthly and in-person strategic decision-making. **All Partners:** ~15% of effort, also spread across the other four WPs, contributing to data collection and ERNEST Day participation.

Coordination and Integration with other WPs

WP5 serves as the pillar for project coordination, linking all WPs and ensuring coherence between scientific, social, and dissemination activities. Coordination mechanisms include: a) **Information flow:** PCT → WPLG → SC for upward reporting; SC → WPLG → PCT for decisions. b) **Periodic ERNEST Days:** Informal yet structured reviews fostering cross-WP integration and early problem detection. c) **Risk and budget monitoring:** Continuous tracking ensures that operational, financial, and social aspects are aligned with project goals. d) **Stakeholder and ERN interaction:** WP5 oversees engagement with external stakeholders and other ERN initiatives, preventing duplication and enhancing project impact.

Table 3.1c: List of Deliverables

#	Deliverable name	Short description	WP	Lead part.	Type	Dissem. level	Delivery month
D5.1	ERNEST Days Notes	Documentation of each regular "ERNEST Days" informal review.	5	INFN	R	PU	1,4,7,10,13, 16,19,22
D5.2	Project Management Handbook	Document detailing the management structure, roles, communication protocols, and conflict resolution mechanisms.	5	INFN	DMP	PU	2
D2.1	Location Plan for ERN events	Detailed mapping of venues where ERNEST will be installed during ERN	2	INFN	R	PU	3,15
D1.1	Promotional toolkit	Promotional materials (posters, flyers, banners, videos, social media kits).	1	UniPg	OTHER	PU	4
D1.2	Project website and online platform	Fully functional project website and active social media accounts (X, Instagram, Facebook, TikTok, LinkedIn).	1	UniPg	DEC	PU	4
D4.1	TOC and Evaluation Framework	Theory of Change (TOC) and Monitoring tools ready for deployment during events, ensuring GDPR compliance.	4	INAF	OTHER	PU	6
D1.3	Training package for researchers	Training content and handouts on science communication and public engagement.	1	UniPg	R	SEN	6,15
D2.2	New Escape Room Scenarios Packages	Materials for newly developed scenarios (scripts, puzzles, props list, setup instr.) on public server (e.g. Zenodo).	2	INFN	DEM	SEN	6,18
D2.3	Scientific Passports Delivery	Personalized digital and physical record showcasing participants' achievements across ERNEST escape rooms.	2	INFN	OTHER	PU	6,18
D3.1	Teacher Training Toolkit	Printable props, story outlines, and challenges linked to school curricula.	3	UniTo	DEM	PU	8
D3.3	Scientific Puzzle Design Contest Guidelines	Official contest rules, communication material, and judging criteria for the puzzle design competition.	3	UniTo	R	PU	15
D1.4	Report on WP1 (year 1)	Summary of Year 1 WP1 activities.	1	UniPg	R	PU	12
D2.4	Report WP2 (year 1)	Documentation of ERNEST WP2 activities for ERN Year 1.	2	INFN	R	PU	12
D3.2	Report WP3 (year 1)	Consolidated report covering all WP3 activities in Year 1.	3	UniTo	R	PU	12
D4.2	Report on WP4 (year 1)	Analysis of data collected during Year 1 activities, with comparison to baseline.	4	INAF	R	PU	12
D5.3	Report on WP5 (year 1)	Formal internal review assessing progress against KPIs, deliverables, milestones, and budget consumption, with recommendations for the second half of the project.	5	INFN	R	PU	12
D1.5	Report on WP1 (year 2)	Two-year overview of campaign results with final impact assessment.	1	UniPg	R	PU	24
D2.5	Report on WP2 (year 2)	Summary of online ERNEST sessions held during ERN Year 2.	2	INFN	R	PU	24

D3.4	Report on WP3 (year 2)	Consolidated report for Year 2 activities, showing improvements, reach expansion, and updated evaluation.	3	UniTo	R	PU	24
D4.3	Report on WP4 (year 2)	Comprehensive report covering both years of activity, with detailed quantitative and qualitative evaluation.	4	INAF	R	PU	24
D5.4	Report on WP5(year 2)	Comprehensive closure document.	5	INFN	R	PU	24

Table 3.1d: List of milestones

#	Milestone name	WPs	Due month	Means of verification
M1.1	Campaign plan and infrastructure ready	1	4	Approved communication plan; promotional toolkit uploaded to shared repository; live website and social media links; distribution plan agreed; all required materials produced, and quality checked.
M1.2	Engagement network activated	1	6	Training attendance lists; training materials uploaded; stakeholder engagement log; signed agreements or confirmations from key partners; outreach calendar.
M1.3	Sustainability pathway agreed	1	24	Network meeting minutes confirming endorsement; draft sustainability plan in repository; letters of intent or other formal commitments from partners for activities.
M2.1	Event locations and scenarios confirmed for year 1 and 2	2	3,15	Final location list for ERN events; scenario descriptions uploaded; signed agreements with local administrations; documented security and safety checks completed.
M2.2	Escape room setups ready for launch for year 1 and 2	2	6,18	Internal readiness checklist; pilot test reports; photo/video documentation of setups; all required items purchased; plots reviewed and approved; dry-run tests successfully conducted with a test audience.
M2.3	Events completed and impact data collected for year 1 and 2	2	12,24	Event reports; participation statistics; summary of feedback forms.
M3.1	Teachers training program designed and launched	3	8	Training curriculum approved; training materials uploaded; schedule and participant list for initial sessions available.
M3.2	First cycle of school activities completed with feedback	3	12	Attendance records; activity reports; contest submissions; feedback forms summarized in report.
M3.3	Final impact report submitted, and sustainability actions approved	3	24	Final impact evaluation report uploaded to repository; meeting minutes or signed document confirming approval of sustainability plan and follow-up activities.
M4.1	Data monitoring system fully operational	4	6	System logs showing active data collection; validation reports; screenshots of dashboard/data streams; documented testing results.
M4.2	Preliminary data analysis and findings reviewed	4	12	Draft preliminary analysis report summarizing data trends and recommendations; presentation slides; consortium meeting minutes recording discussion, feedback, and decisions on next steps.
M4.3	Comprehensive impact analysis and evaluation report completed	4	24	Formal approval document confirming endorsement of the final evaluation report and decision to submit to the EU.
M5.1	Management procedures approved	5	3	Official consortium meeting minutes recording approval of management procedures.

M5.2	Mid-term project and risk review done	5	12	Meeting minutes documenting review outcomes and agreed corrective actions.
M5.3	Project formally closed and lessons learned endorsed	5	24	Meeting minutes or signed declaration confirming formal project closure and approval of final lessons learned report.

Table 3.1e: Critical risks for implementation #@RSK-MGT-RM@#

Description of risk (indicate level of (i) likelihood, and (ii) severity: Low/Medium/High)	**WPs**	**Proposed risk-mitigation measures**
Delays in finalizing and approving campaign materials lead to missed launch deadlines for digital presence and outreach ((i)Low, (ii) Low)	1	Set internal deadlines ahead of official ones; conduct early peer reviews; assign dedicated communication leads; use agile iterative feedback cycles.
Low attendance or dropout in researcher/tutor training sessions reduces quality and coverage of engagement activities ((i)Low, (ii) Medium)	1	Partner with academic institutions for recruitment; offer multiple sessions/time slots; provide incentives and certificates; monitor signup trends and adapt recruitment.
Failure to secure formal agreements with local authorities causes last-minute cancellation or relocation of escape room events ((i)Low, (ii) High)	2	Start agreements at least 6 months prior; maintain ongoing dialogue; prepare fallback venues; document safety and legal compliance early; assign local liaison officers.
Late procurement or missing essential materials delays setup and dry-run testing of escape rooms, compromising quality ((i)Medium, (ii) Medium)	2	Prepare detailed procurement timeline; pre-approve suppliers with proven reliability; maintain buffer stock; conduct early dry runs with substitute materials; track procurement weekly; prioritize using sellers with established trustworthiness and good track records to minimize supply risks.
Low teacher engagement or limited uptake of training programs reduces the scale and effectiveness of school activities. ((i)Low, (ii) Medium)	3	Establish strong partnerships with educational institutions and authorities, starting with schools already within our established network; design engaging and adaptable training modules; recognize participant achievements through certifications; continuously track enrollment patterns and adjust recruitment strategies as needed.
Technical glitches or GDPR non-compliance in online questionnaires reduce data quantity and quality for impact assessment ((i)Medium, (ii) High)	4	Use GDPR-compliant platforms verified before launch; test extensively with sample users; have manual backup data collection; appoint data protection officer.
Inconsistent or incomplete data collection leads to unreliable baseline and impact measures. Delays in preliminary data analysis postpone consortium decision-making and affect final reporting timeline ((i)Medium, (ii) High)	4	Train event staff on data collection protocols; implement real-time data completeness monitoring; perform spot checks; revise tools promptly based on feedback. Allocate dedicated analysis resources early; set intermediate deadlines with reminders; use collaborative tools for transparency; schedule interim analysis reviews.
Public health restrictions or sudden political changes force cancellation or shift of planned in-person events ((i)Low, (ii) High)	1,2,3,4,5	Develop hybrid event formats (in-person + online); maintain flexible contracts with venues; monitor local regulations; have communication plan ready for stakeholders.
Potential overlap or competition between this project and existing flagship ERN events conducted by the same institution may lead to inefficiencies due to duplicated efforts or audience segmentation. ((i)Low, (ii) High)	1,2,3,4,5	Highlight that ERNEST offers a complementary and innovative engagement format through immersive escape rooms, designed specifically to reach audiences and locations often underserved by traditional ERN activities. Emphasize coordinated planning and joint communication with existing ERN initiatives to ensure synergy and avoid duplication. This strategic collaboration maximizes outreach impact, broadens audience diversity, and supports the EU's goal of

		inclusive and innovative science engagement.

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Table 3.1f: Summary of staff effort

	WP1	WP2	WP3	WP4	WP5	Total PM/Participant
BE1: INFN	1.5	**2.5**	1	1	**3.5**	9.5
BE2: INAF	0.5	1	0.5	**1**	0.5	3.5
BE3: CNR	0.5	1.5	1	0.5	0.5	4
BE4: CNRS	0	0.5	0.5	0	0	1
BE5: UniBa	0	1	1	0.5	0	2.5
BE6: UniPi	0	0.5	0.5	0	0	1
BE7: UniPv	0	1	0.5	0	0	1.5
BE8: UniPg	**1.5**	0.5	0.5	0	0.5	3
BE9: UniMan	0.5	2	1.5	0	0	4
BE10: UniUPO	0	0.5	0.5	0	0	1
BE11: UniTo	0.5	0.5	**1.5**	0.5	0.5	3.5
BE12: LAMA	1	0	0	4	0.5	5.5
Total PM	6	**11.5**	**9**	**7.5**	**6**	**40**

Table 3.1h: 'Purchase costs' items (travel and subsistence, equipment and other goods, works and services)

BE1: INFN	**Cost (€)**	**Justification**
Travel and subsistence	4.5k	Travel to ERN location and schools
Other goods, works and services	27.2k	Realization of new escape rooms/kits and material for activities as schools. Gadgets and visual identity material for the entire consortium for 2 years.
Total	31.7k	
BE2: INAF	**Cost (€)**	**Justification**
Travel and subsistence	4k	Travel to ERN location and schools
Other goods, works and services	7k	Realization of new escape rooms/kits and material for activities at schools.
Total	11k	
BE3: CNR	**Cost (€)**	**Justification**
Travel and subsistence	1k	Travel to ERN location and schools
Other goods, works and services	8k	Realization of new escape rooms/kits and material for activities as schools.
Total	9k	
BE4: CNRS	**Cost (€)**	**Justification**
Travel and subsistence	2k	Travel to ERN location and schools

Other goods, works and services	4k	Realization of new escape rooms/kits and material for activities as schools.
Total	6k	
BE5: UniBa	**Cost (€)**	**Justification**
Other goods, works and services	10k	Realization of new escape rooms/kits and material for activities as schools.
Total	10k	
BE6: UniPi	**Cost (€)**	**Justification**
Travel and subsistence	1k	Travel to ERN location and schools
Other goods, works and services	10k	Realization of new escape rooms/kits and material for activities as schools.
Total	11k	
BE7: UniPv	**Cost (€)**	**Justification**
Other goods, works and services	3k	Realization of new escape rooms/kits and material for activities as schools.
Total	3k	
BE8: UniPg	**Cost (€)**	**Justification**
Travel and subsistence	2k	Travel to ERN location and schools
Other goods, works and services	11k	Realization of new escape rooms/kits and material for activities as schools. Total budget for website realization and maintenance.
Total	13k	
BE9: UniMan	**Cost (€)**	**Justification**
Travel and subsistence	3.6k	Travel to ERN location and schools
Other goods, works and services	6.7k	Realization of new escape rooms/kits and material for activities as schools. Total budget for website realization and maintenance.
Total	10.3k	
BE10: UniUPO	**Cost (€)**	**Justification**
Travel and subsistence	2k	Travel to ERN location and schools
Other goods, works and services	3k	Realization of new escape rooms/kits and material for activities as schools.
Total	5k	
BE11: UniTo	**Cost (€)**	**Justification**
Travel and subsistence	2.5k	Travel to ERN location and schools
Other goods, works and services	10k	Realization of new escape rooms/kits and material for activities as schools.
Total	12.5k	

3.2 Capacity of participants and consortium as a whole #@CON-SOR-CS@# #@PRJ-MGT-PM@#

The ERNEST consortium has been deliberately built to combine **diversity, geographical reach, and complementary expertise**, ensuring that the project's objectives can be achieved with efficiency, inclusivity, and long-term sustainability. Its structure reflects three strategic criteria that represent its major strengths.

1. Strong participation of Italian national research institutions with a distributed presence
Key national research organizations—**INFN, CNR, and INAF**—play a central role in ERNEST through several departments spread across the Italian territory, from North to South and both major islands. This widespread structure guarantees that the project can reach citizens in multiple regions, offering equal opportunities for participation regardless of geographical location. The distributed model also enables each node to r**eplicate and adapt escape-room kits** locally, reducing costs and maximizing reach. By exploiting their infrastructures and networks, these institutions ensure high scientific quality and operational scalability. **INFN** is led by **Livia Soffi and Francesca Cavallari**, inventor of **HEPscape! (winner of SIF outreach prize in 2024)** and previous coordinators of major physics and detectors groups within the CMS collaboration at CERN. They will be also the **leaders of WP 3 and 5**. They won also an internal call for outreach activities in Sapienza University, Rome, together with their PhD students to realize HEPscape! in Italian sign language (https://www.phys.uniroma1.it/it/insegniamo-le-particelle). The INFN enters the consortium with 4 departments (Roma, Torino, Catania and Bari). The team includes **Simone Mastrogiovanni,** ERC-St grantee 2024 and high-profile expert of Gravitational Waves and **Carlo Nicolau and Mariangela Bondì**, active members of the KM3NET collaboration, to ensure a solid expertise for the development of the novel escape room scenarios suggested in section 1.2.4.3. **CNR** group is represented by **Maria Francesca Carfora (IAC) and Alessandra Rocco (INO)**. The group is part of the network **CNR CREO (Campania Region Outreach)** and includes representatives from 8 CNR Institutes, from biology to chemistry, physics, mathematics, engineering. This wide range of expertise has already led to the realization of the multidisciplinary Escape Room **"Alla ricerca dell'equilibrio perduto"** and will be instrumental in developing the new scenarios suggested in Section 1.2.4.3. The **INAF** team is led by **Claudia Mignone** and **Maura Sandri**, astrophysicists and science communicators, who have extensive experience with creating, presenting and evaluating astronomy educational resources using coding and robotics, including the acclaimed escape room **"A bordo di un fotone"**, and by **Serena Pastore**, expert evaluator of European projects, especially in the framework of Erasmus+ and Horizon Europe programs, who is currently **delegate for INAF in the European Open Science Cloud association**. They will also be **leaders of WP4**. INAF enters the consortium with 6 of its premises (Padova, OAS Bologna, Roma HQ, Torino, Cagliari, IASF Palermo). The team also includes **Silvia Casu, radio astronomer** with expertise in astrochemistry, author of the **"A caccia nel buio escape room"**, who is a member of the operational team of the Sardinia Radio Telescope (SRT) and outreach contact point for the SETI (Search for Extraterrestrial Intelligence) project in Italy, to ensure a solid expertise for the development of the novel escape room scenarios suggested in section 1.2.4.3.

2. Active involvement of universities and specialized partners
The universities participating in ERNEST contribute a rich diversity of expertise—from science communication and education to multilingual learning and teacher training. They also represent a wide and dynamic audience base, engaging thousands of students, educators, and families in the co-creation and dissemination of the project's activities. Complementing the academic and research members, **LAMA Società Cooperativa** (led by **Mattia Forni)** brings a unique and "orthogonal" competence. LAMA is a Social Sciences and Humanities partner specialized in **social impact evaluation**. With extensive experience in assessing educational poverty and community-based learning projects, LAMA will lead the design of the project's impact framework and contribute to the analysis and reporting activities. Its interdisciplinary team combines expertise in evaluation, participatory research, and social innovation, ensuring methodological rigor and relevance to societal challenges. **UniPi** (led by **S. Donato)** and **UniPv** (led by **P. Montagna)** strengthen ERNEST through HEPscape! And "PER me si va", recently brought for the first time to **Lucca Comics & Games, Europe's largest comics festival**. As ERNEST's reference hub for Tuscany, Lombardy and Liguria, Pisa and Pavia connect with the research institutes INFN and CNR and with schools of excellence Scuola Normale Superiore and Scuola Superiore Sant'Anna. Pisa's proximity to the European Gravitational Observatory (EGO, Cascina) enables the co-design and testing of innovative physics-based scenarios. **UniTo** is led by **Anna Vignati**, professor and researcher of the Medical Physics group and coordinator of the Third Mission Committee of the UniTo Physics Dept and winner of a special mention in the SIF Outreach prizes at SIF in 2024 and inventor of the **Medical Physics escape room**. Together with **Elisa Corino (Foreign Languages Dept,** scientific coordinator UNIGHT project, horizon-msca-2023-citizens-01), they are inventors and key drivers of the "**maSTEM Game escape-room"** initiative combining **STEM content and CLIL** (Content and Language Integrated Learning) methods. The UniTo team will **lead WP3** and includes **experts in methodologies and teaching technologies in Physics (Marta Rinaudo** and **Antonio Amoroso)**, to ensure a solid expertise for the development of the activities foreseen in WP3. The team from **UniUPO is represented by Marta**

Ruspa, Professor of Applied Physics at the School of Medicine, and Nicolò Jacazio, tenure-track researcher. Both are members of LHC experiments (CMS and ALICE, respectively). Marta Ruspa has long served as coordinator of the Department's Third Mission Committee and has been actively involved in numerous outreach initiatives and collaborative projects with local schools in the Novara, Vercelli, and Verbano-Cusio-Ossola areas. The team has already collaborated in previous years with the group from UniTo and will work together with them on the development of portable kits to be proposed to schools across Piedmont. **UniBa** (led by **Federica Simone**) has recently seen its Physics Department recognized as a "Department of Excellence" by the Italian Ministry of University and Research for the **QUASIMODO project (Quantum Sensing and Modeling for One-Health)**. It is considered a national reference in quantum science and technology, with advanced research in quantum optics and sensing and the interuniversity Master's Program in Quantum Science and Technology (MaSTeR-Quantum). Through ERNEST the university aims to integrate a **new escape room on Quantum Technologies** (see 1.2.4.3) into its consolidated outreach activities, and in cooperation with the local INFN department. **UniPg** (led by **V. Mariani, outreach and dissemination responsible of the INFN department in Perugia and co-leader of HEPscape!**) is ideally positioned to lead WP1 thanks to its proven excellence in science communication and large-scale public engagement program. UniPg recently **won the "Polaris" call** – Innovative initiatives for the orientation and training of students in STEM disciplines, confirming its leadership in designing impactful educational campaigns. As one of Italy's most active universities in outreach and institutional communication, UniPg has established strong, long-term relationships with schools, cultural networks, municipalities, and regional media outlets. Its communication team has successfully contributed to awareness campaigns for national and European initiatives, combining traditional media, digital platforms, and territorial partnerships to reach diverse audiences, including underserved communities.

3. International participation and cross-border exchange

ERNEST brings together partners from different European countries, proposing a multicultural and interdisciplinary environment that enhances the project's creativity and impact. This diversity allows typical national formats of science engagement to be exported, adapted, and enriched across borders. By sharing methodologies and co-developing new escape-room models, the partners will "contaminate" each other with innovative practices, promoting a genuinely European dimension of collaboration and mutual learning. The **UniMan** team is led by **Morgan Hibberd** and **Caterina Doglioni** from the Department of Physics & Astronomy at The University of Manchester. Morgan Hibberd holds an STFC Ernest Rutherford Fellowship (ERF 2023) and is an expert in laser and terahertz techniques for advanced particle acceleration and photonics. He is the **outreach leader for the Cockcroft Institute** - the national center for accelerator R&D - a partnership of STFC and the universities of Manchester, Lancaster, Liverpool and Strathclyde, located at Daresbury National Laboratory. He is creator of the particle accelerator escape room "**Acceler8 Escape"** and has previously worked on the award-winning (European Physical Society Outreach Award 2019) outreach project **Tactile Collider**, which made the physics of accelerators **accessible to the visually impaired** through tactile objects, binaural sounds and braille workbooks. This will underpin the development of a unique "tactile" escape room, which UniMan will co-create with the visually impaired community (e.g. specialist teachers, students) as part of WP2 and WP3 to spread the ERNEST values of inclusivity and accessibility. Caterina Doglioni (ERC-St grantee 2015, ERC-CoG grantee 2020) is a dark matter and real-time analysis expert working within the ATLAS collaboration. She holds several coordination roles in European grants - most relevant to this project is the coordination of the SMARTHEP training network where the team of Early-Stage Researchers have **translated the HEPscape! room to 5 different languages** and made the instructions for the escape room fully FAIR and available on Zenodo. Plans have been outlined for a new real-time analysis (trigger) based escape room (see sec.1.2.4.3), which UniMan will continue developing during ERNEST in WP2 with the summer interns supported by this project. Through the university, institute, national lab and funding council networks, the UniMan team **provides a well-connected UK escape-room hub** to deliver innovation as part of the ERNEST program. The **CNRS** team is led by **Emilie Maurice and Christophe Thiebaux**, physicists at the laboratory. Both work on the LHC experiments (LHCb and CMS), assuming significant responsibilities. Christophe Thiebaux has led the laboratory's communication and outreach activities for 10 years. Emilie Maurice is a professor at the École Polytechnique and heads the "Fundamental Laws of the Universe" pole in the Physics Department. Both organize local sessions of international particle physics masterclasses. They will be assisted by **Sylvaine Pieyre, an experienced graphic designer**. A key outreach partner will be the **Pole Égalité des Chances** of the École Polytechnique, with whom CNRS has an established collaboration through laboratory visits, internships and career orientation activities. The project will further benefit from additional access to schools through the existing **IN2P3 educational network**. For broader dissemination and public visibility, CNRS will collaborate with the **press and communication services of École Polytechnique, IN2P3**, which provides direct channels to both general and science-focused national media.

Together, these features make ERNEST a cohesive yet diverse consortium, capable of combining **top-level scientific expertise with inclusive outreach and methodological innovation**. Each participant has a well-defined role and adequate resources to fulfil it, contributing to a collective capacity that matches the project's objectives and the ambition to create a sustainable, European network of science engagement through educational escape rooms. #§CON-SOR-CS§# #§PRJ-MGT-PM§# #§QUA-LIT-QL§# #§WRK-PLA-WP§#